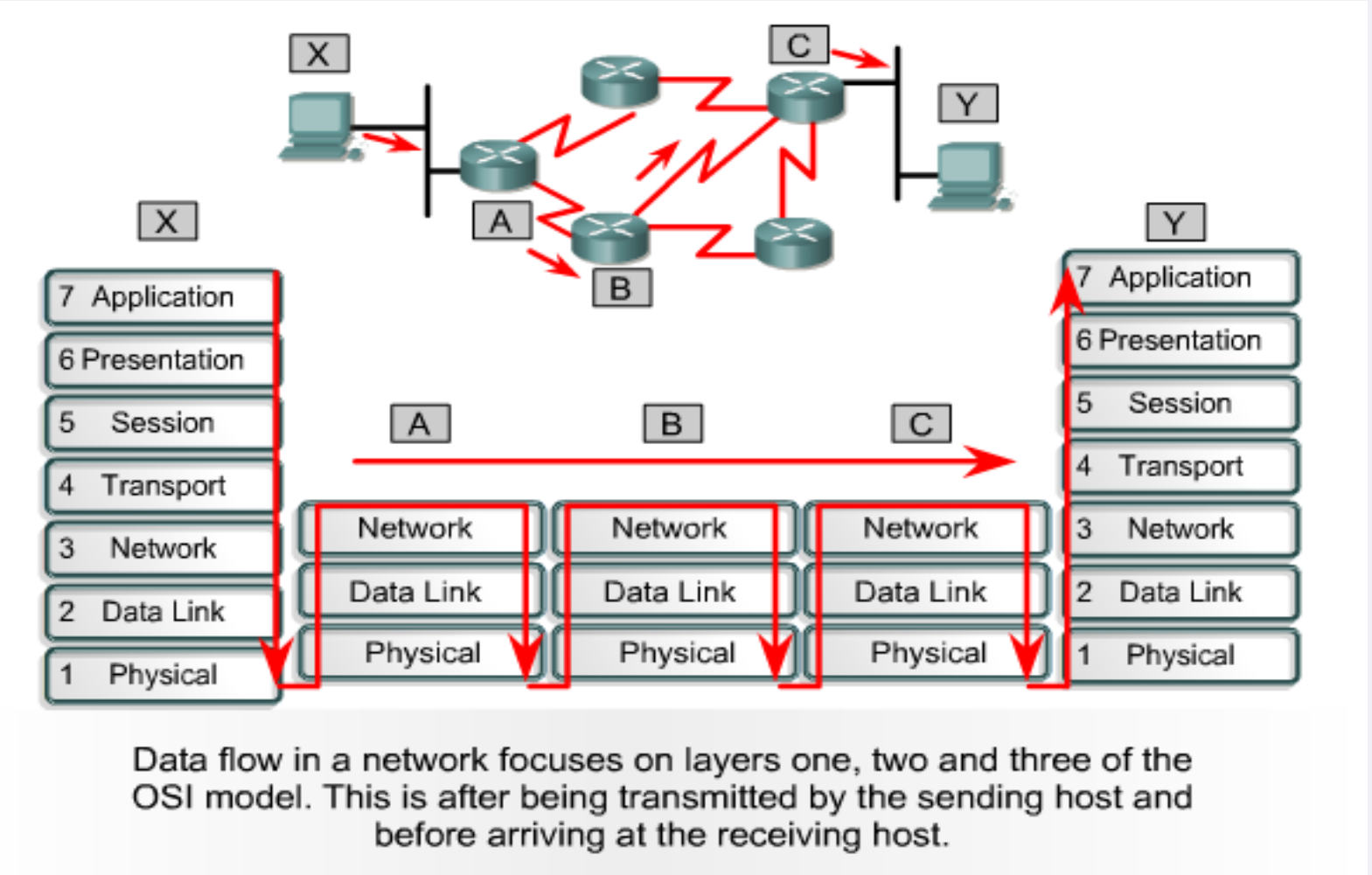
Switched Network Overview

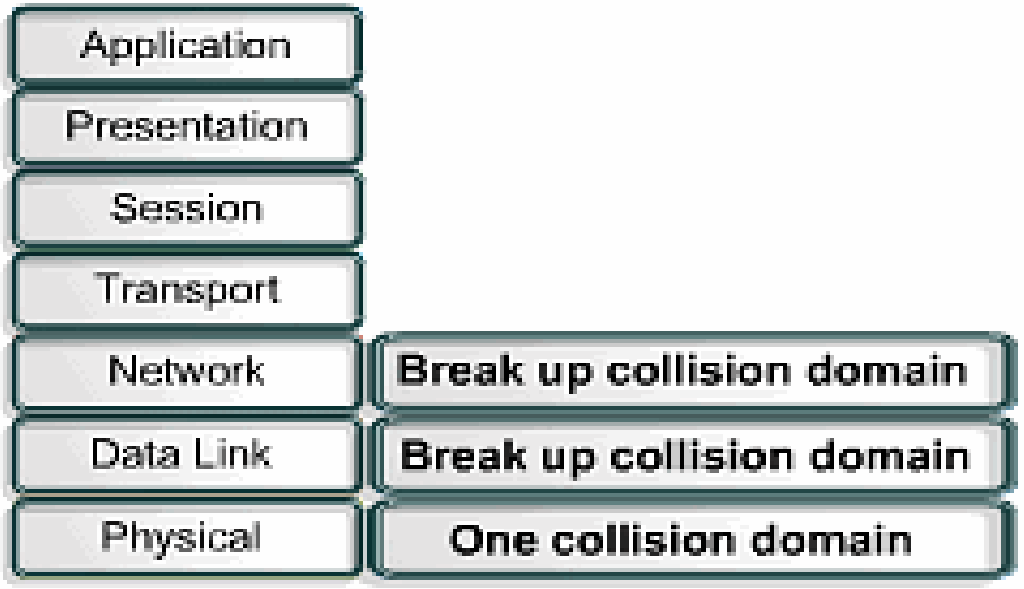
Instructor: Auttapon Pomsathit: CCNA

Data Flow Through a Network

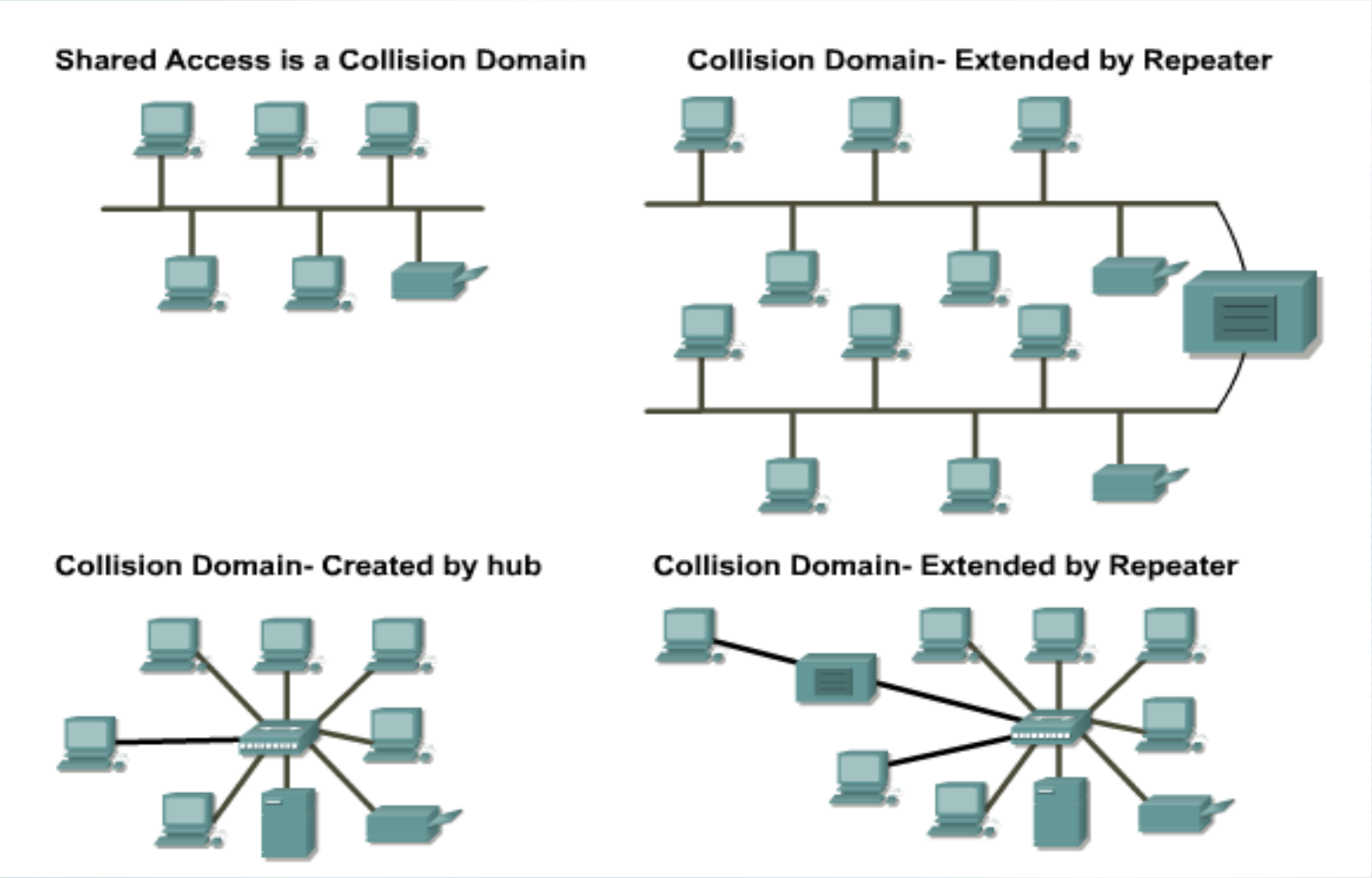


Data flow in a network focuses on layers one, two and three of the OSI model. This is after being transmitted by the sending host and before arriving at the receiving host.

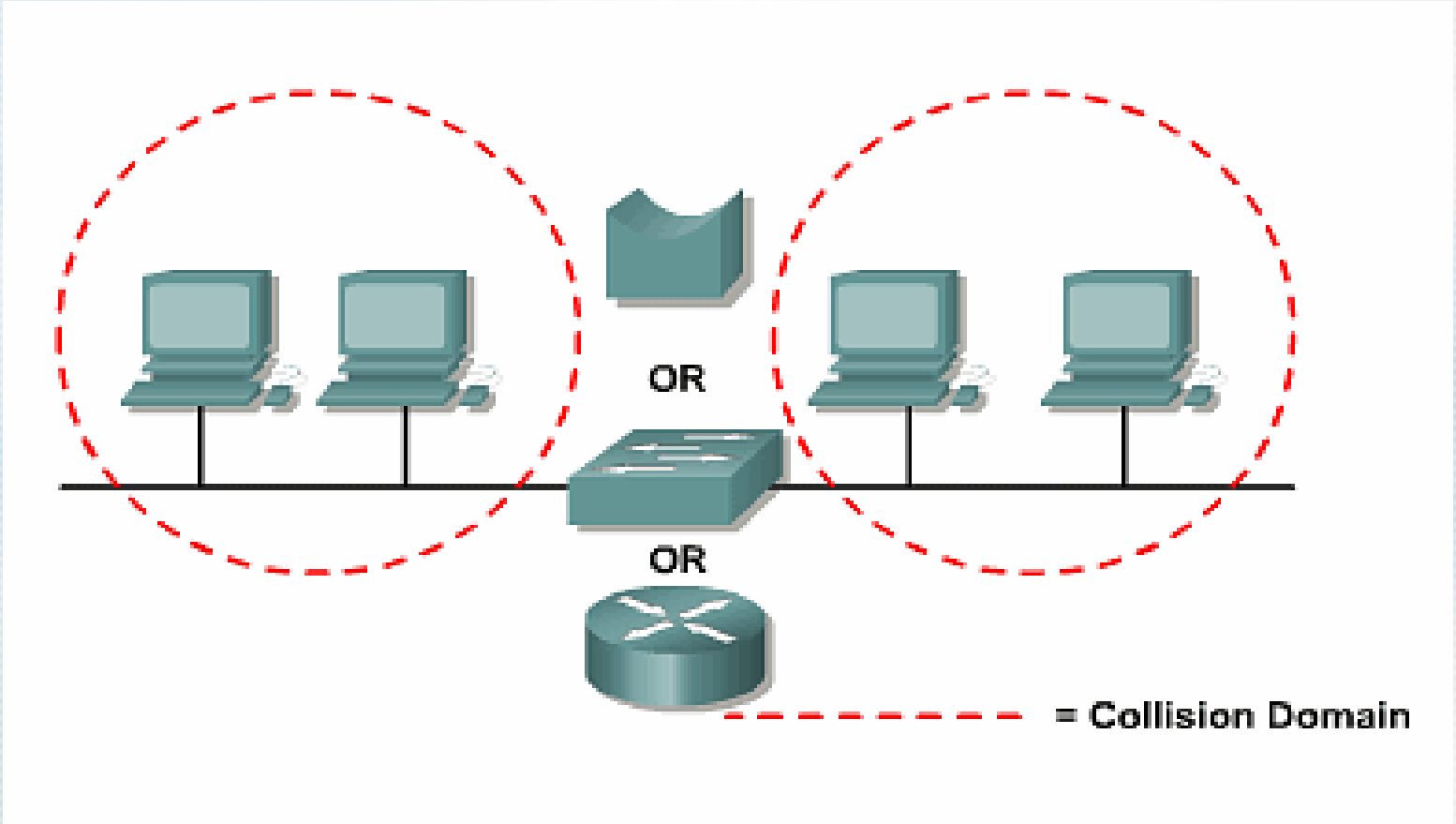
Collision Domain Segmentation



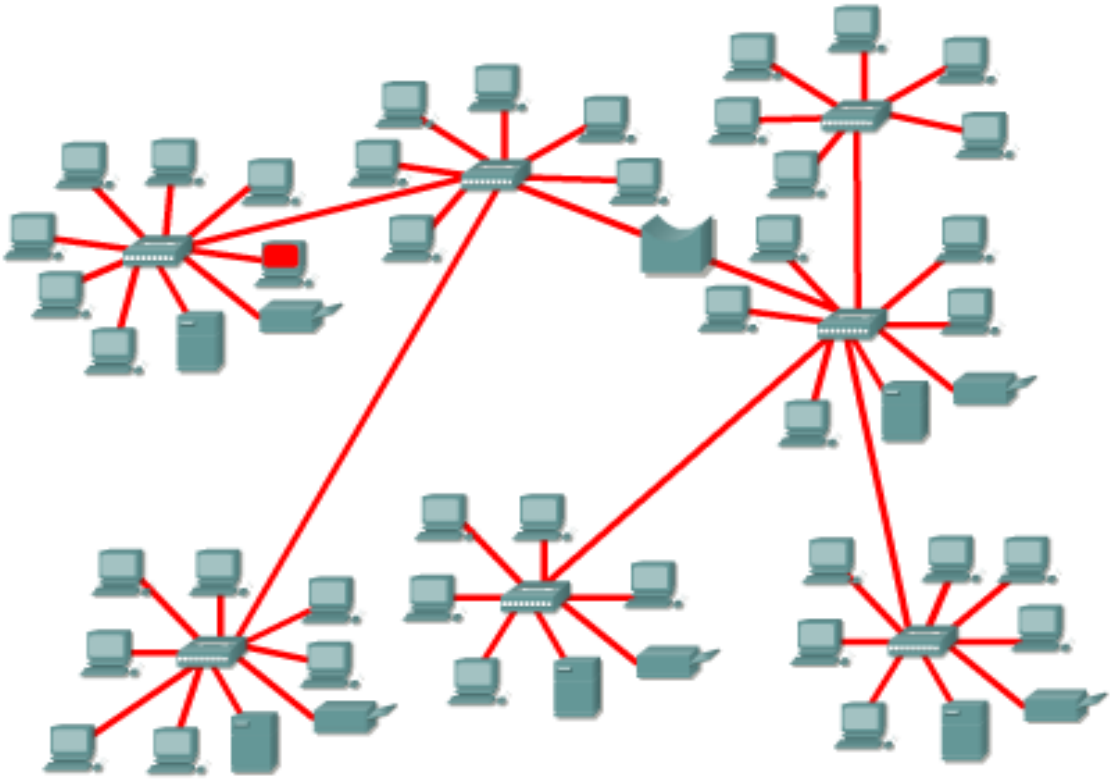
Layer 1 Devices Extend Collision Domains



Limiting the Collision Domains

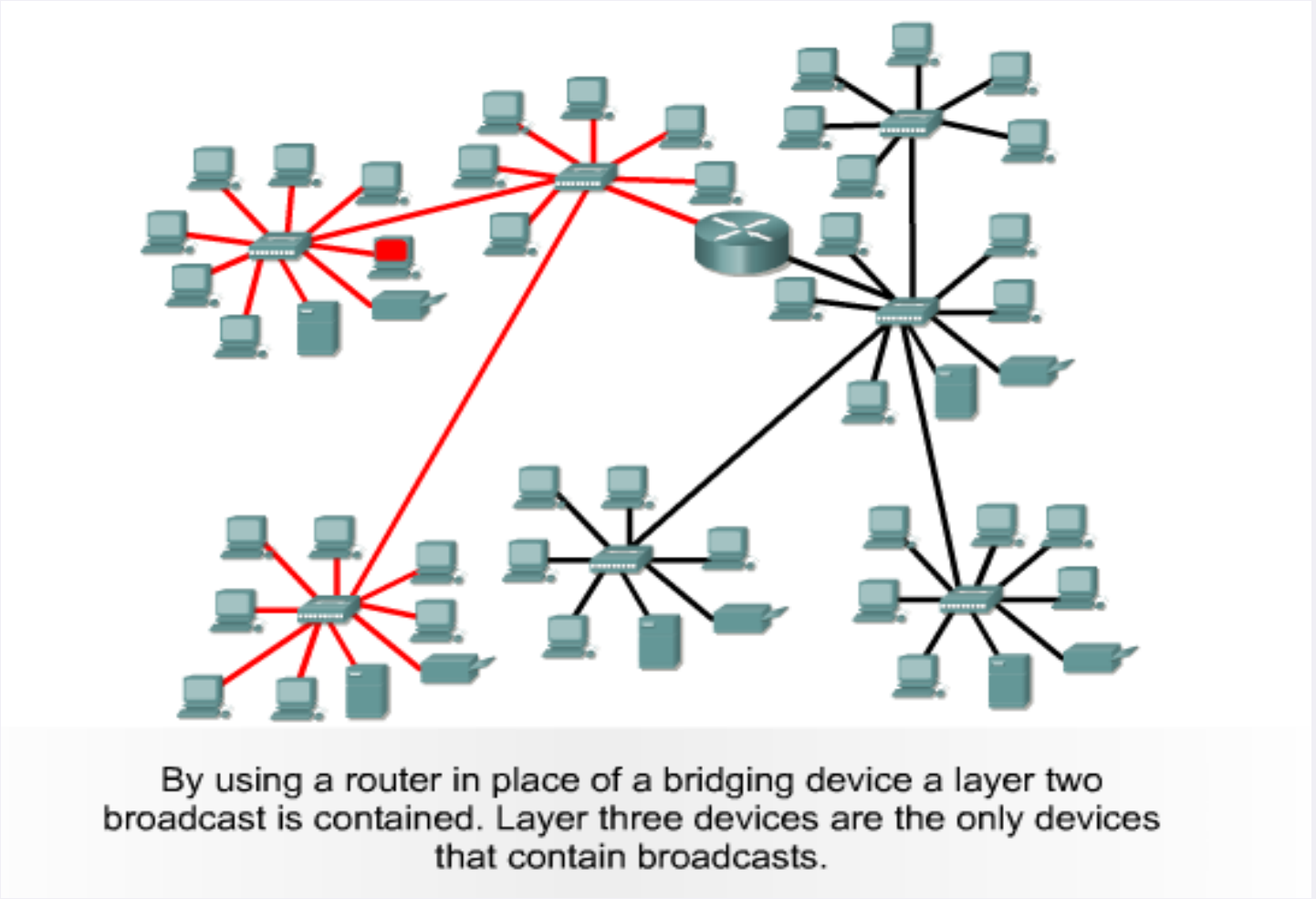


Broadcasts in a Bridged Environment



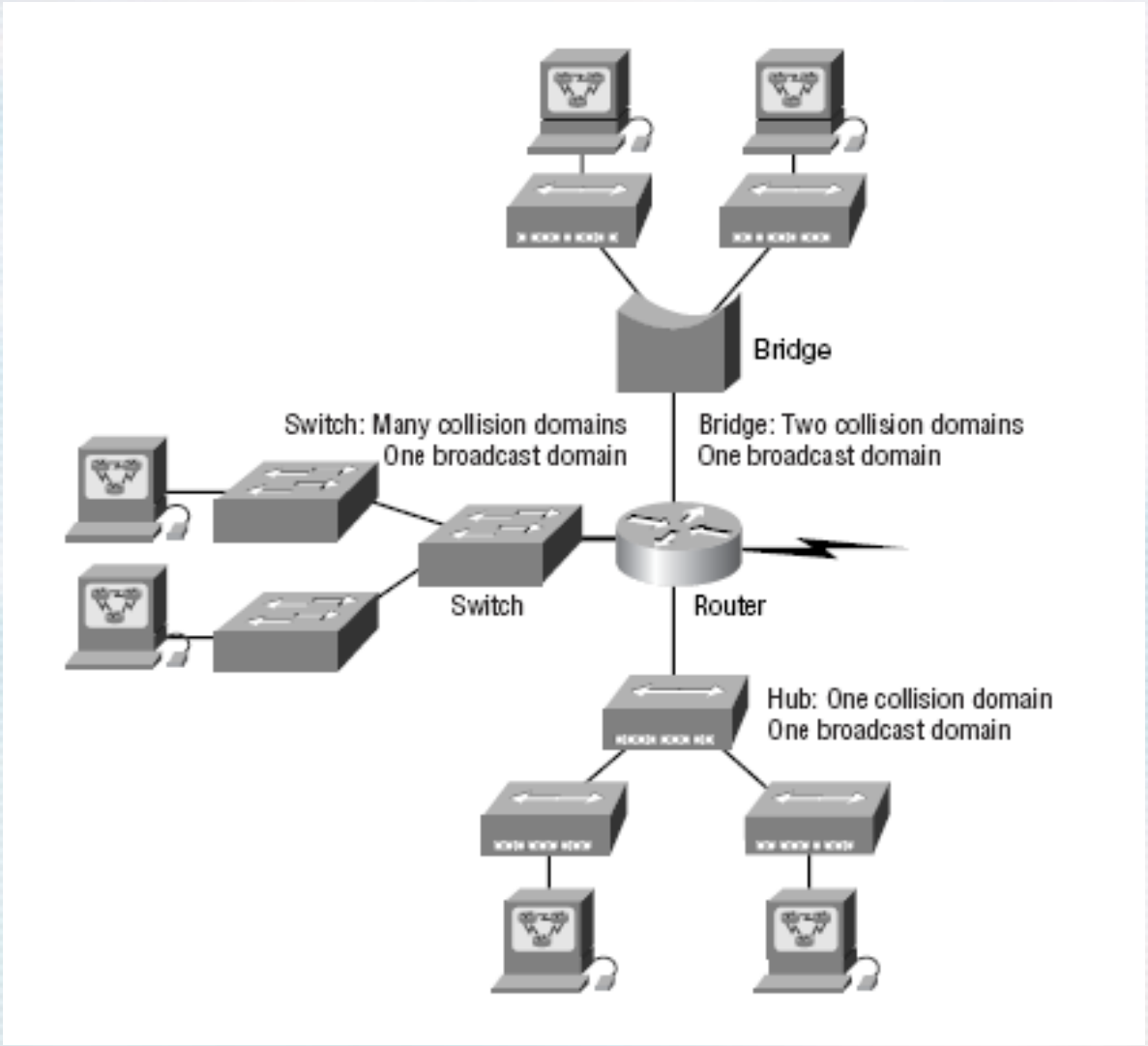
A broadcast is picked up by all stations. A broadcast is also forwarded across all bridges whether the receiving host is on the other side of the bridge or not. This eliminates the benefits of having a bridged network.

Broadcast Domain Segmentation



By using a router in place of a bridging device a layer two broadcast is contained. Layer three devices are the only devices that contain broadcasts.

Collision Domain and Broadcast Domain



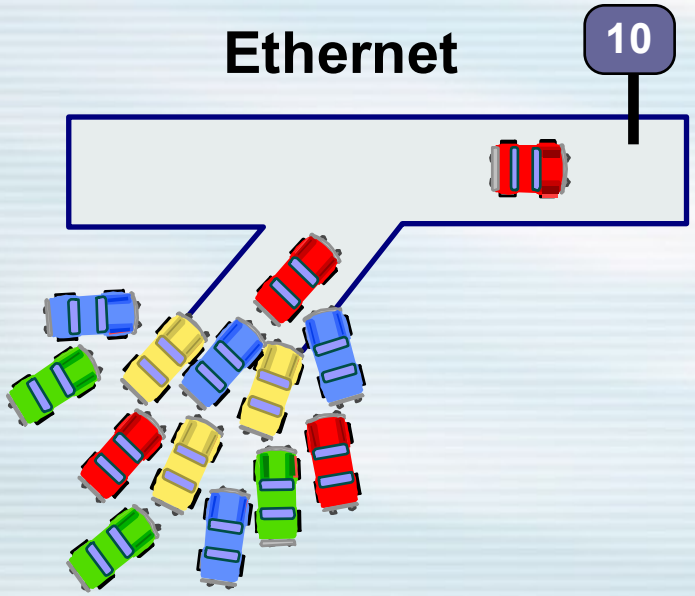
Summary

Summary

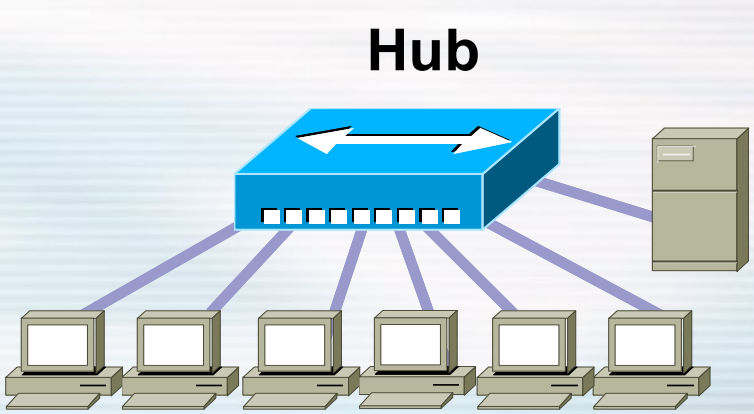
An understanding of the following key points should have been achieved:

- Common switching modes:
  - Store-and-forward
  - Cut-through
  - Fragment-free
- Spanning-Tree Protocol is used to resolve and eliminate loops in a network.
- Layer 1 devices do not break up collision domains.
- Layer 2 and Layer 3 devices break up collision domains.
- Layer 1 and Layer 2 devices do not block broadcasts.

Hubs Addressed Many of These Problems



One device sending at a time

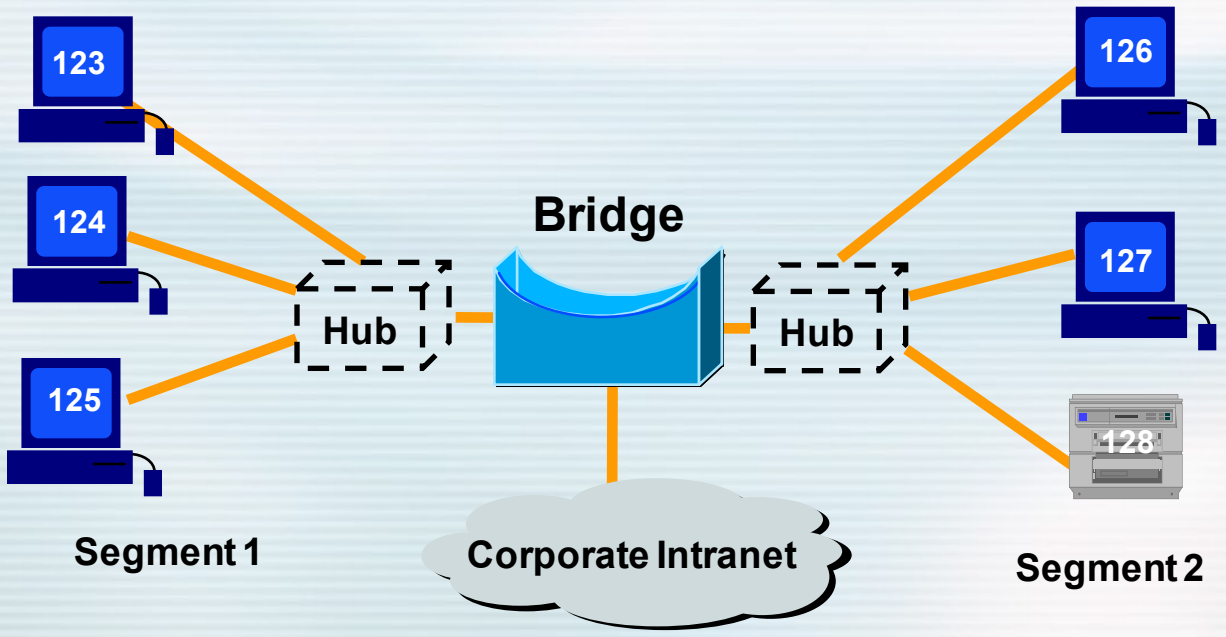


All nodes share 10 Mbps

- Ethernet concentrator
- "Self-contained" Ethernet LAN in a box
- Passive
- Works at physical layer 1

Bridges

- More intelligent than a hub
- “Eavesdrop” on conversations to learn and maintain address tables
- Collect and pass packets between two network segments
- Control traffic to the network

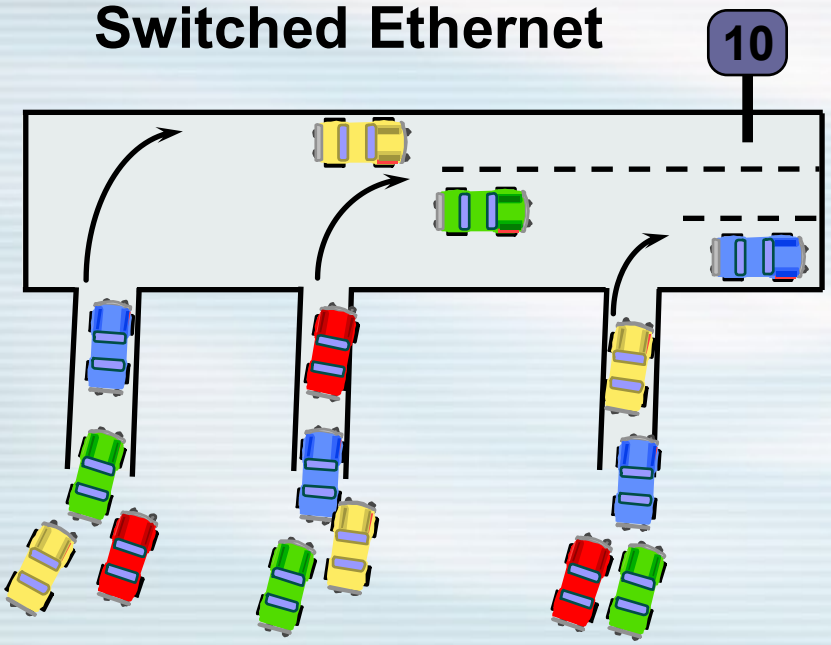


Switching

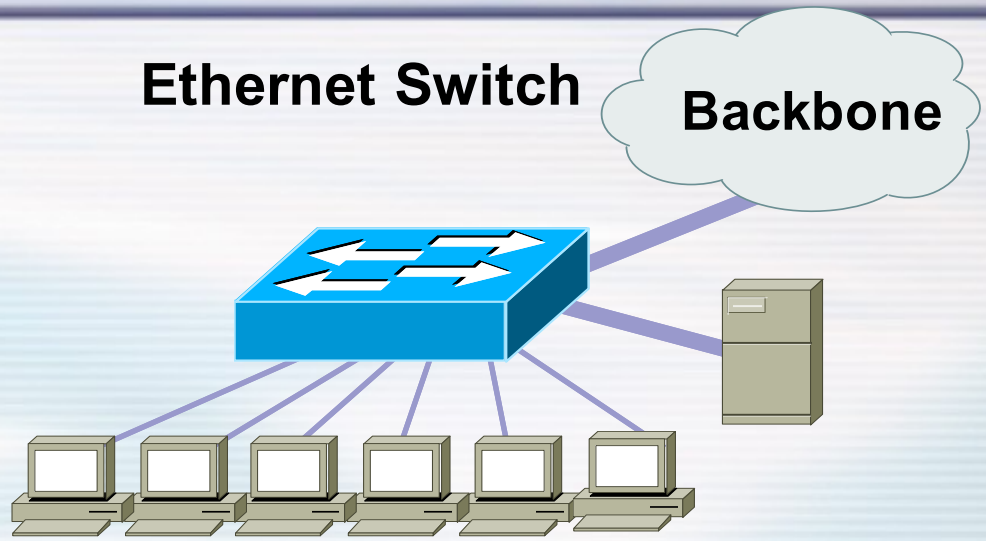
- switches enable high speed data exchange.
- a switch cannot translate from one media type to another.
- switches work in hardware using ASICs.
- The main advantage of switches is that they ease administration of the network.
- A marginal increase in latency is the only disadvantage of LAN switching.
- three major functions of a switch are
    - packet forwarding/filtering.
    - broadcast filtering, and address learning.

loop avoidance using STP

Switches—Layer 2

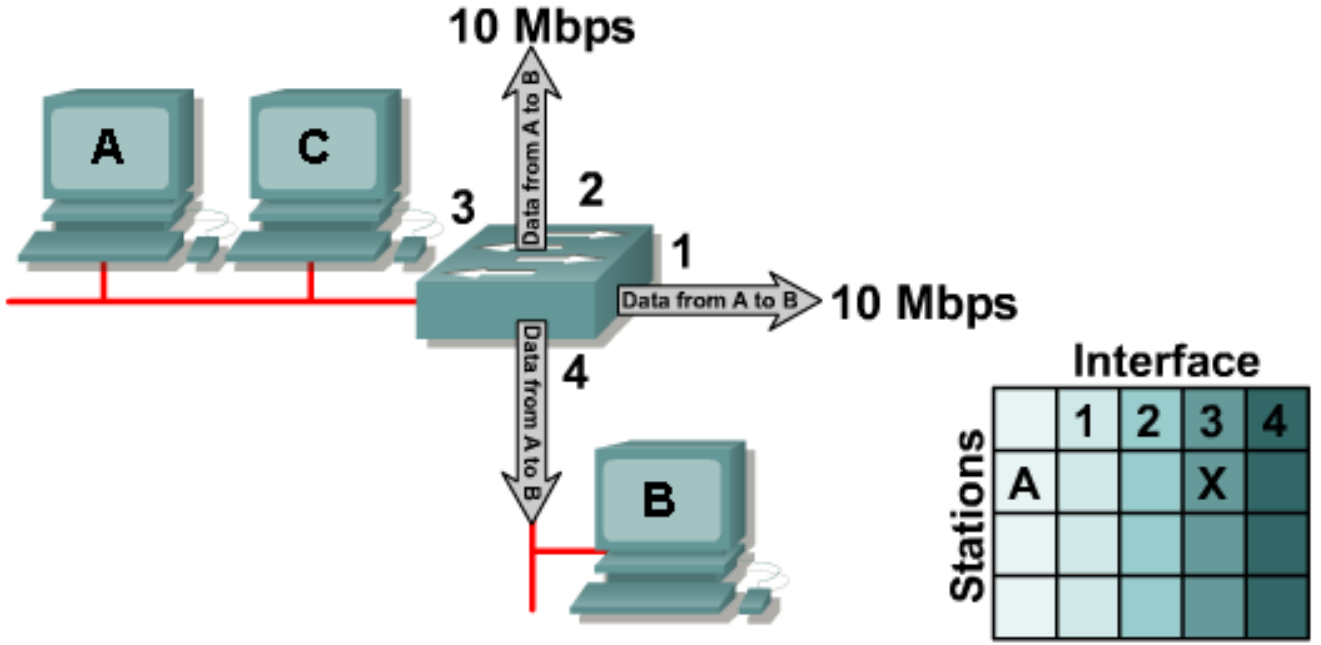


Multiple devices sending at the same time



Each Node has
10 Mbps

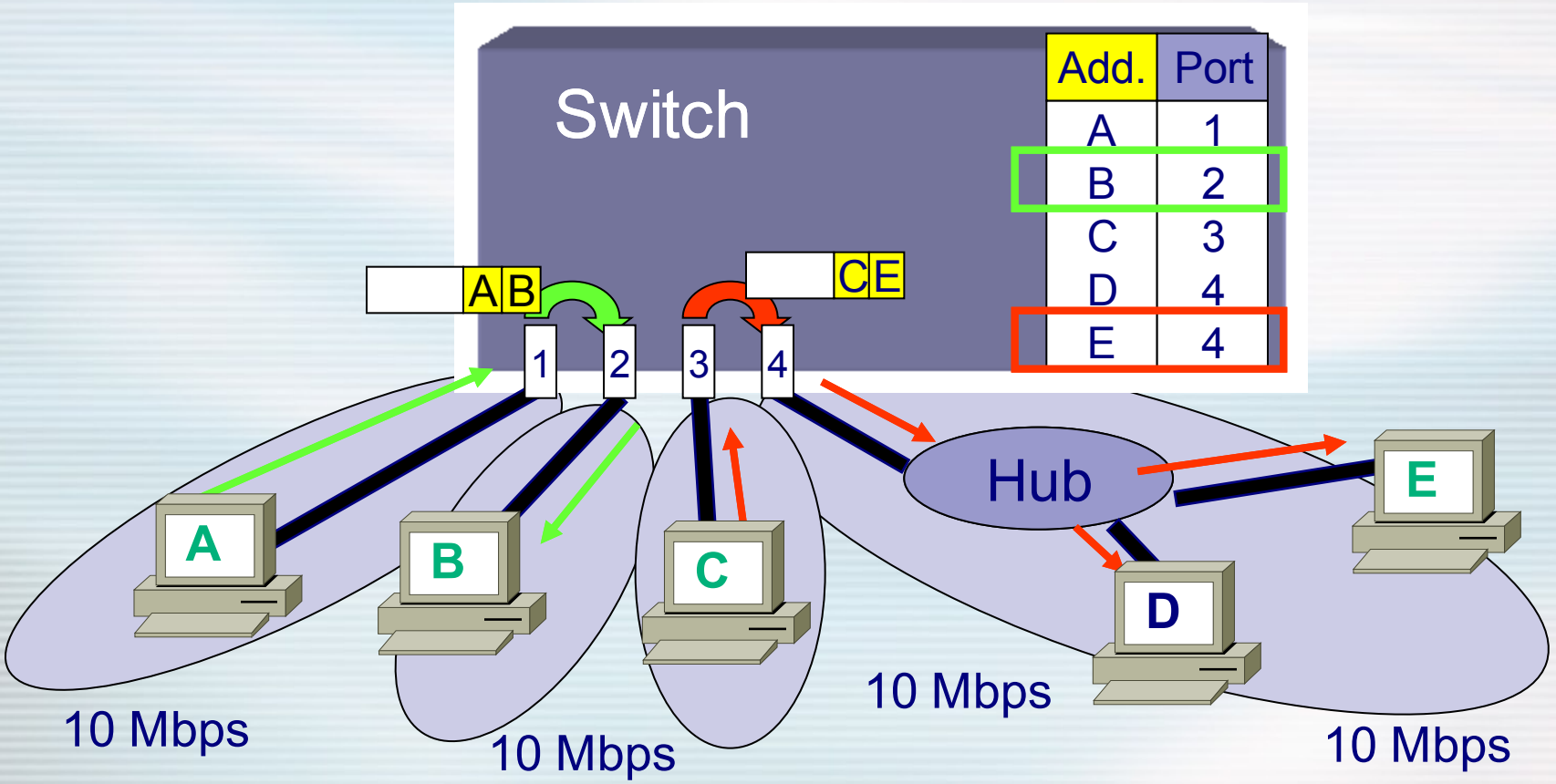
Switch Operation



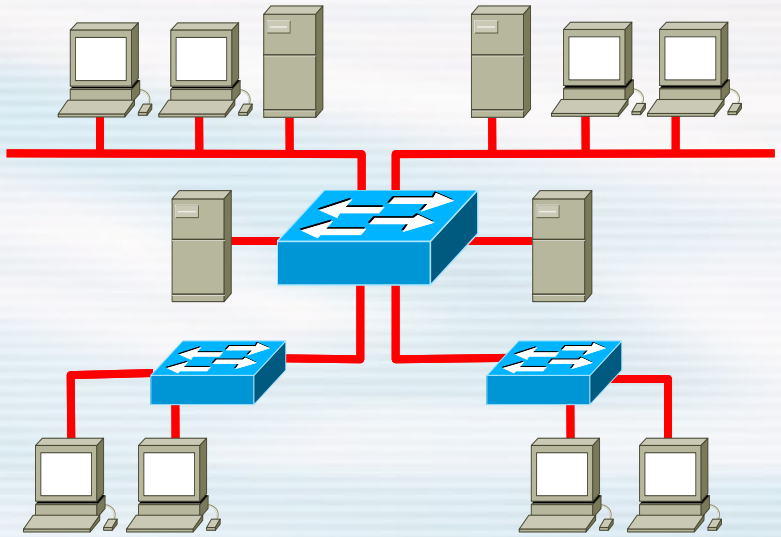
- Forward packets based on MAC address in forwarding table
- Operates at OSI Layer 2
- Learns a station's location by examining source address

Switched Ethernet

Switched Ethernet is a high performance multi-port bridge

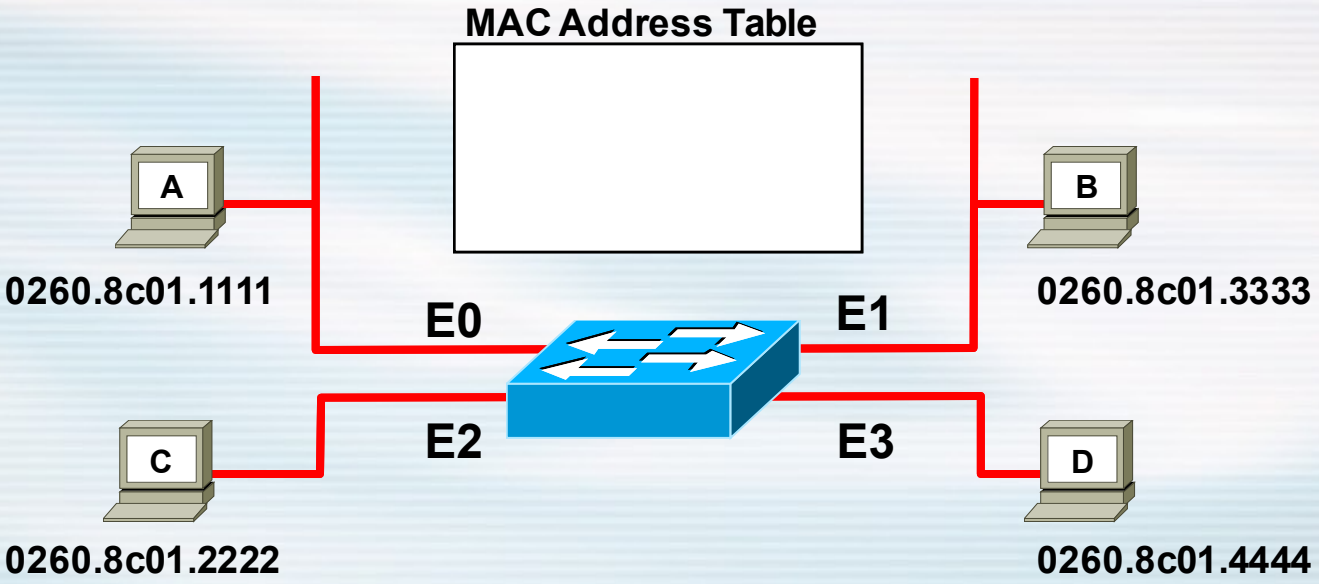


Three Switch Functions



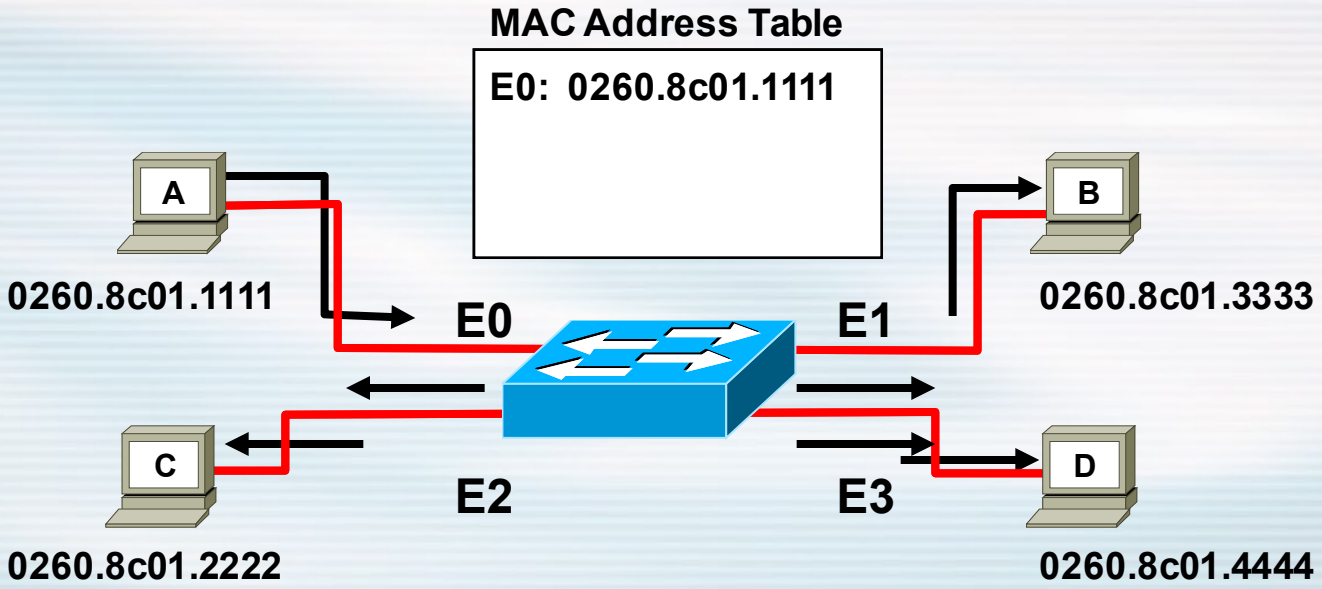
- Address learning
- Forward/filter decision
- Loop avoidance

How Switches Learn Host Locations



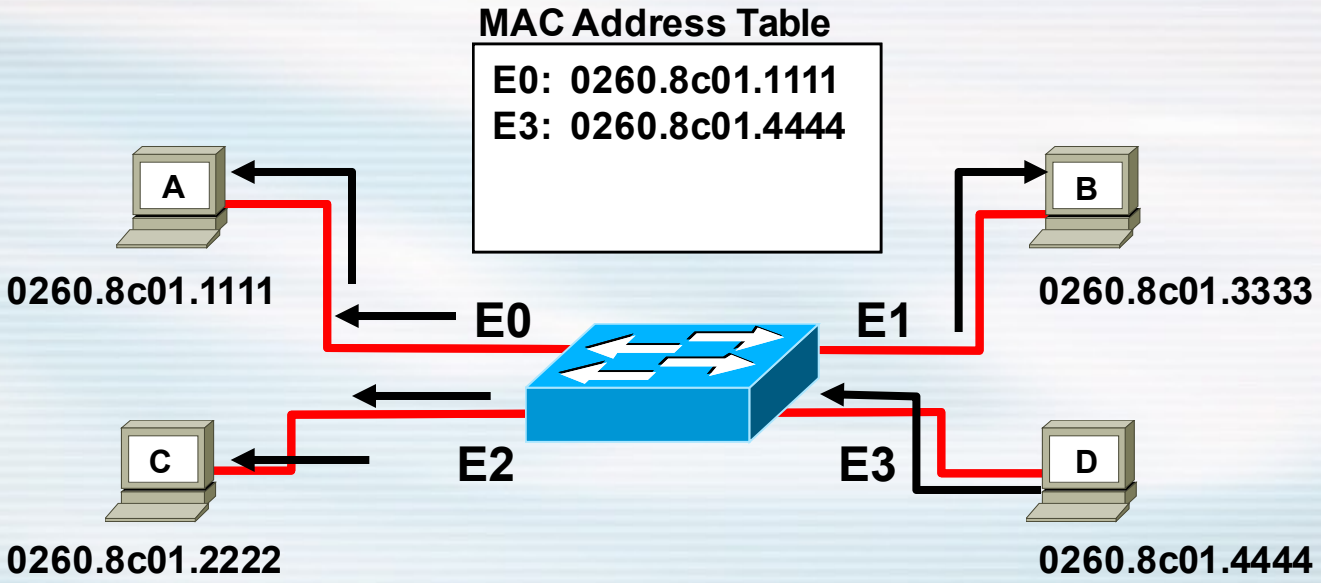
- **Initial MAC address table is empty.**

How Switches Learn Host Locations



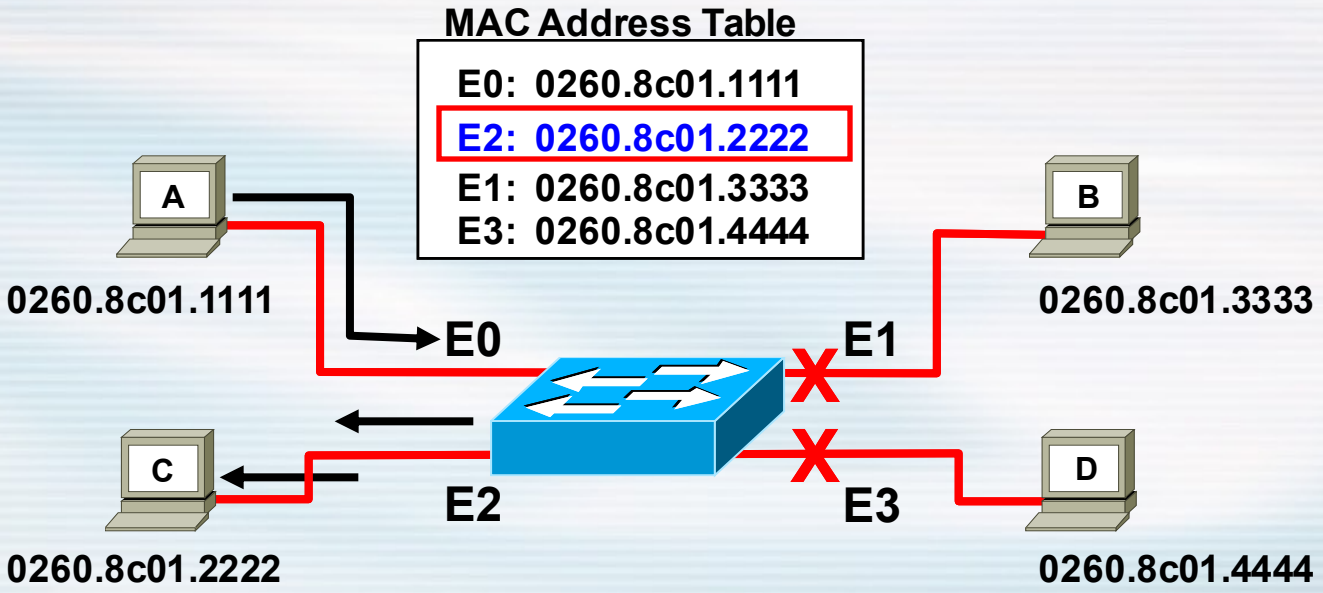
- Station A sends a frame to station C.
- Switch caches the station A MAC address to port E0 by learning the source address of data frames.
- The frame from station A to station C is flooded out to all ports except port E0 (unknown unicasts are flooded).

How Switches Learn Host Locations



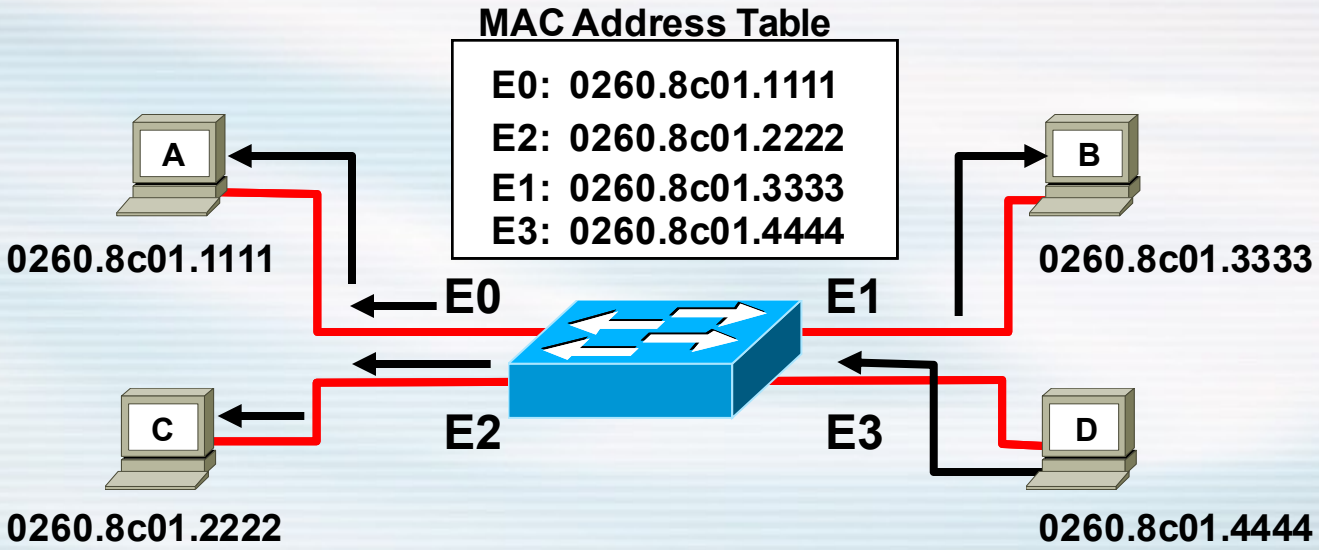
- Station D sends a frame to station C.
- Switch caches the station D MAC address to port E3 by learning the source address of data frames.
- The frame from station D to station C is flooded out to all ports except port E3 (unknown unicasts are flooded).

How Switches Filter Frames



- Station A sends a frame to station C.
- Destination is known; frame is not flooded.

Broadcast and Multicast Frames



- Station D sends a broadcast or multicast frame.
- Broadcast and multicast frames are flooded to all ports other than the originating port.

Managing the MAC Address Table on Catalyst Switch

```
Switch#show mac-address-table
Dynamic Address Count:                  2
Secure Address Count:
Static Address (User-defined) Cou
System Self Address Count:
Total MAC addresses:
Maximum MAC addresses:
Non-static Address Table:
Destination Address  Address Type
Port
-------------------  ------------
0010.7a60.ad7e       Dynamic
00e0.2917.1884       Dynamic
```

```
Switch#clear mac-address-table
Switch#show mac-address-table
Dynamic Address Count:                  0
Secure Address Count:                   0
Static Address (User-defined) Count:    0
System Self Address Count:              13
Total MAC addresses:                    14
Maximum MAC addresses:                  8192
Non-static Address Table:
Destination Address  Address Type  VLAN  Destination
Port
-------------------  ------------  ----  ---------------
```

- The MAC address table is also referred to as the **CAM table (Content Address Memory)** on some switches.
- The default MAC address table aging time is **300 seconds**.

Transmitting Frames Through a Switch

- **Cut-Through**

copies only destination address (6 bytes) into buffer

the frame is forwarded once the destination port is known

no error checking

low & constant rate of latency



- **Fragment-Free (Modified Cut-Through)—Cat1900 Default**

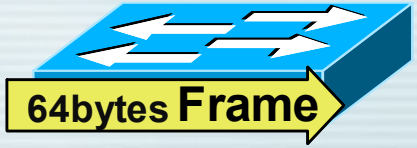
Does not forward the packet until the data portion is received (first 64 bytes).

If the frame is < 64 bytes, then the switch discards the frame, as Ethernet specifications state that collisions should be detected during the first 64 bytes of a frame.

Most collisions are therefore filtered, but late collisions are still possible.

The latest Ethernet switch (1900) products default to Fragment-Free.

- .



- **Store and Forward**

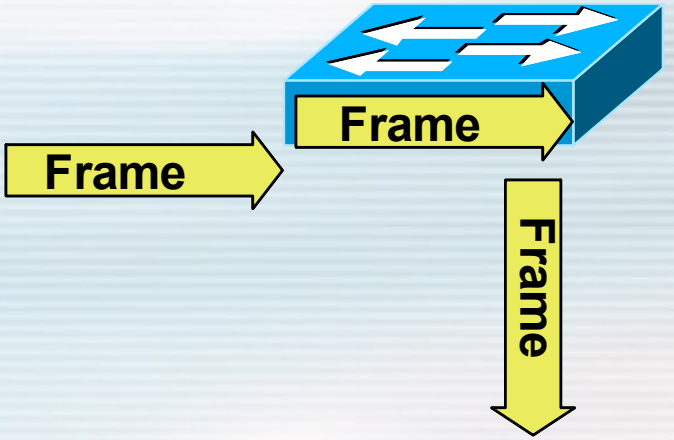
Store-and-Forward

copies entire frame into buffers and checks CRC

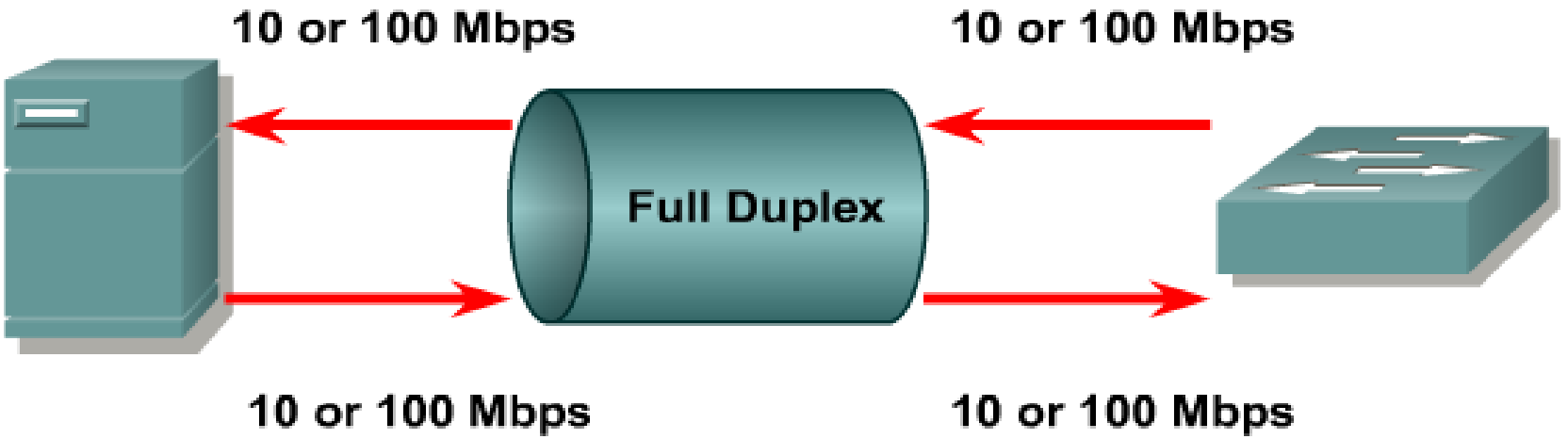
If check is good, MAC table is consulted and the frame is forwarded to the appropriate port

Good error checking

High & variable rate latency (depending on frame size)



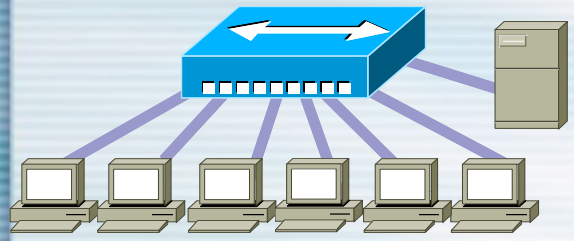
Full-Duplex Transmitting



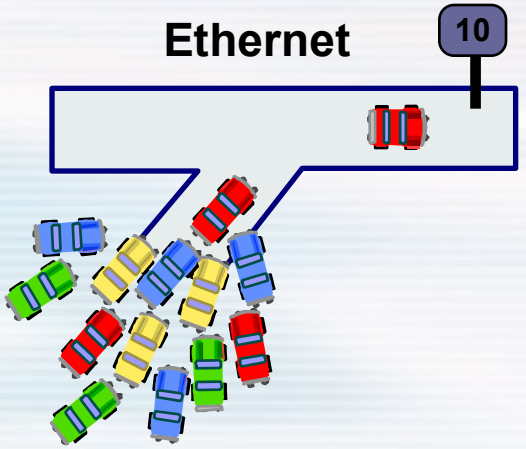
- Doubles bandwidth between nodes (for example, switch and server)
- Collision-free transmission
- Two 10-or 100-Mbps data paths

Switches versus Hubs

Hub

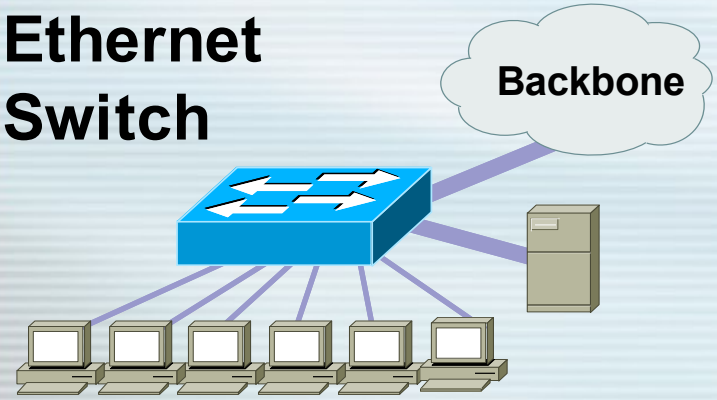


All nodes share 10 Mbps

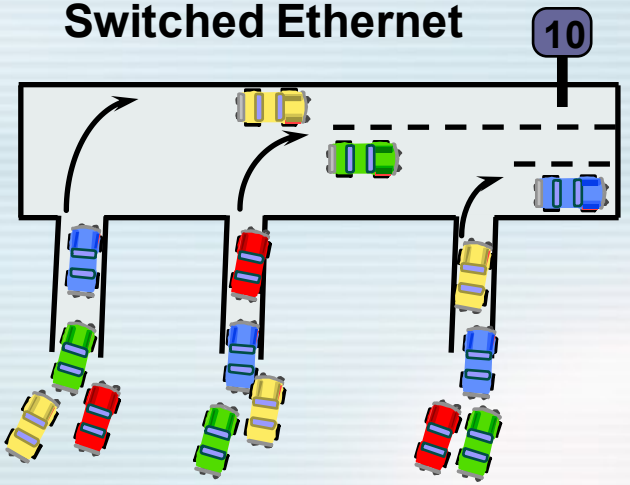


One device sending at a time

Ethernet Switch



Each node has 10 Mbps



Multiple devices sending at the same time

Bridging Compared with LAN Switching

Bridging



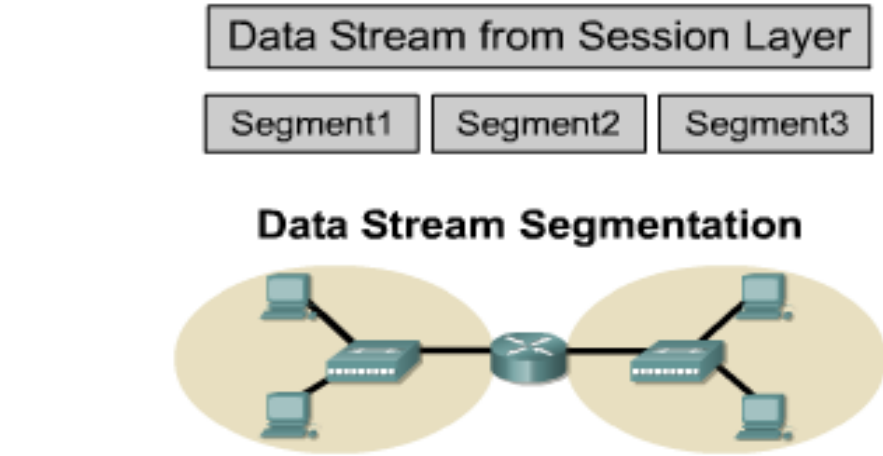
- Primarily software based
- One spanning-tree instance per bridge
- Usually up to 16 ports per bridge

LAN Switching



- Primarily hardware-based (ASIC)
- Many spanning-tree instances per switch
- More ports on a switch

Segments



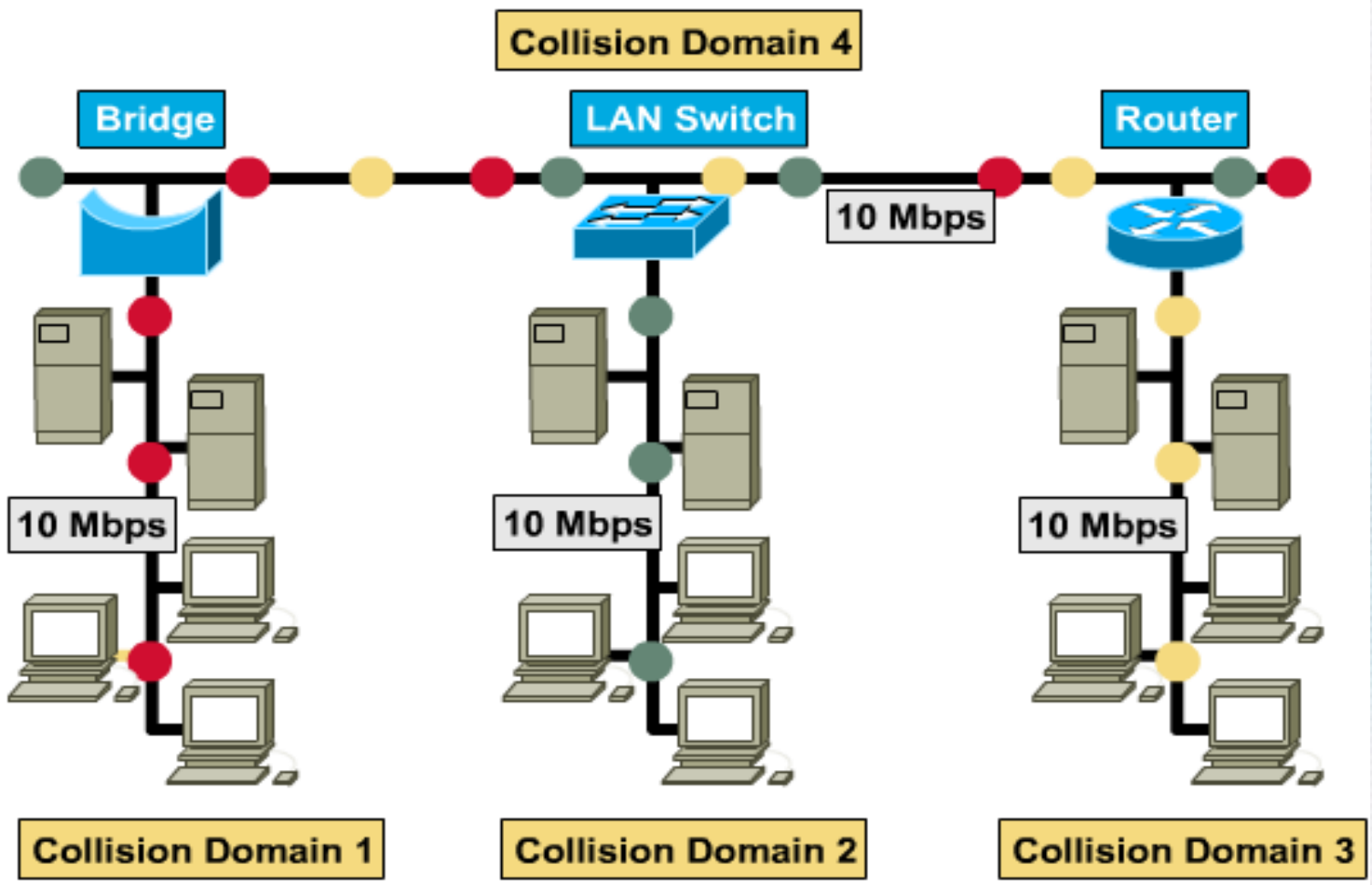
Two Network Segments



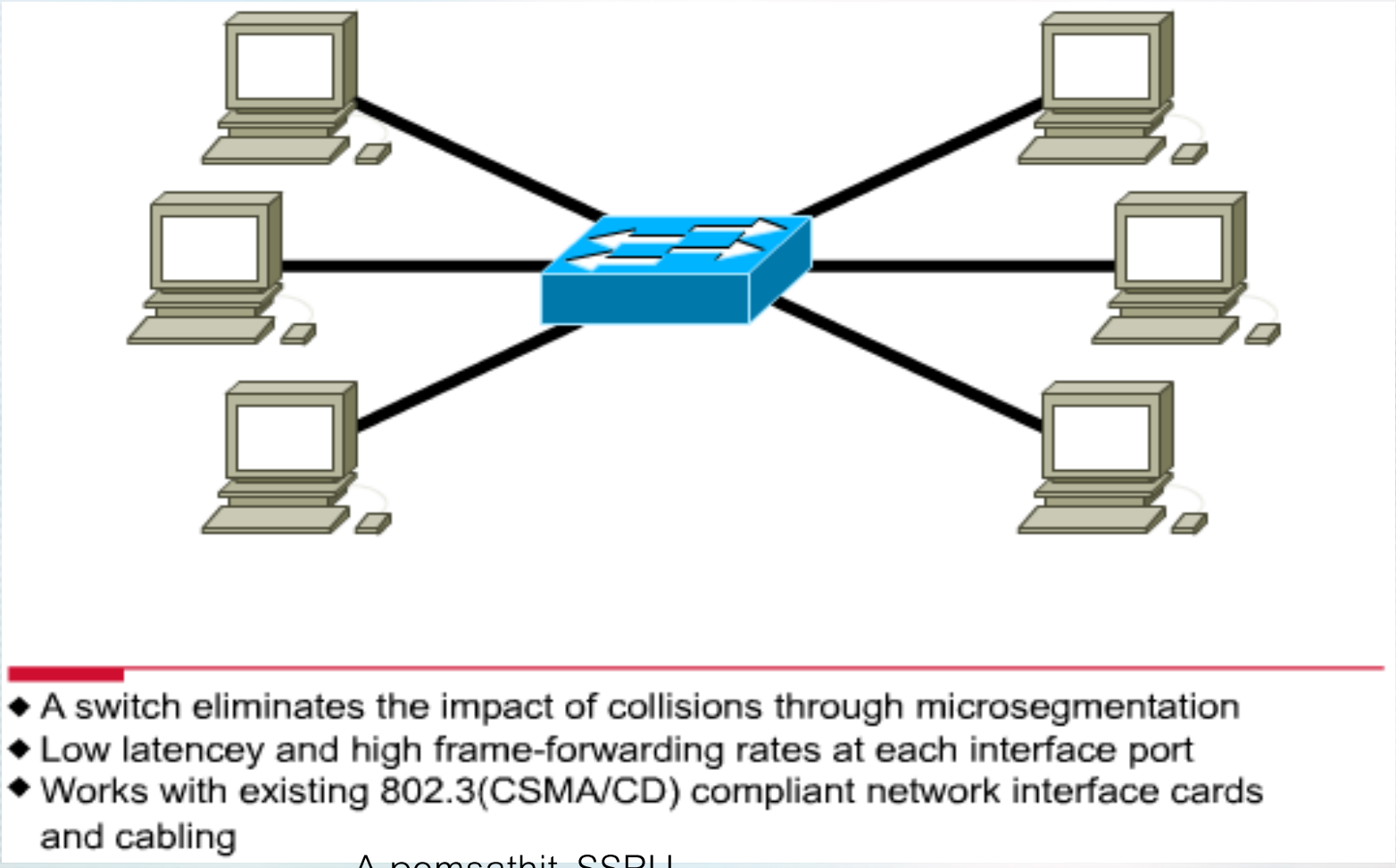
Wire Segment

There are different types of segments in networking. The meaning of the term **segments** depends on the context of a sentence.

Ethernet LAN Segmentation



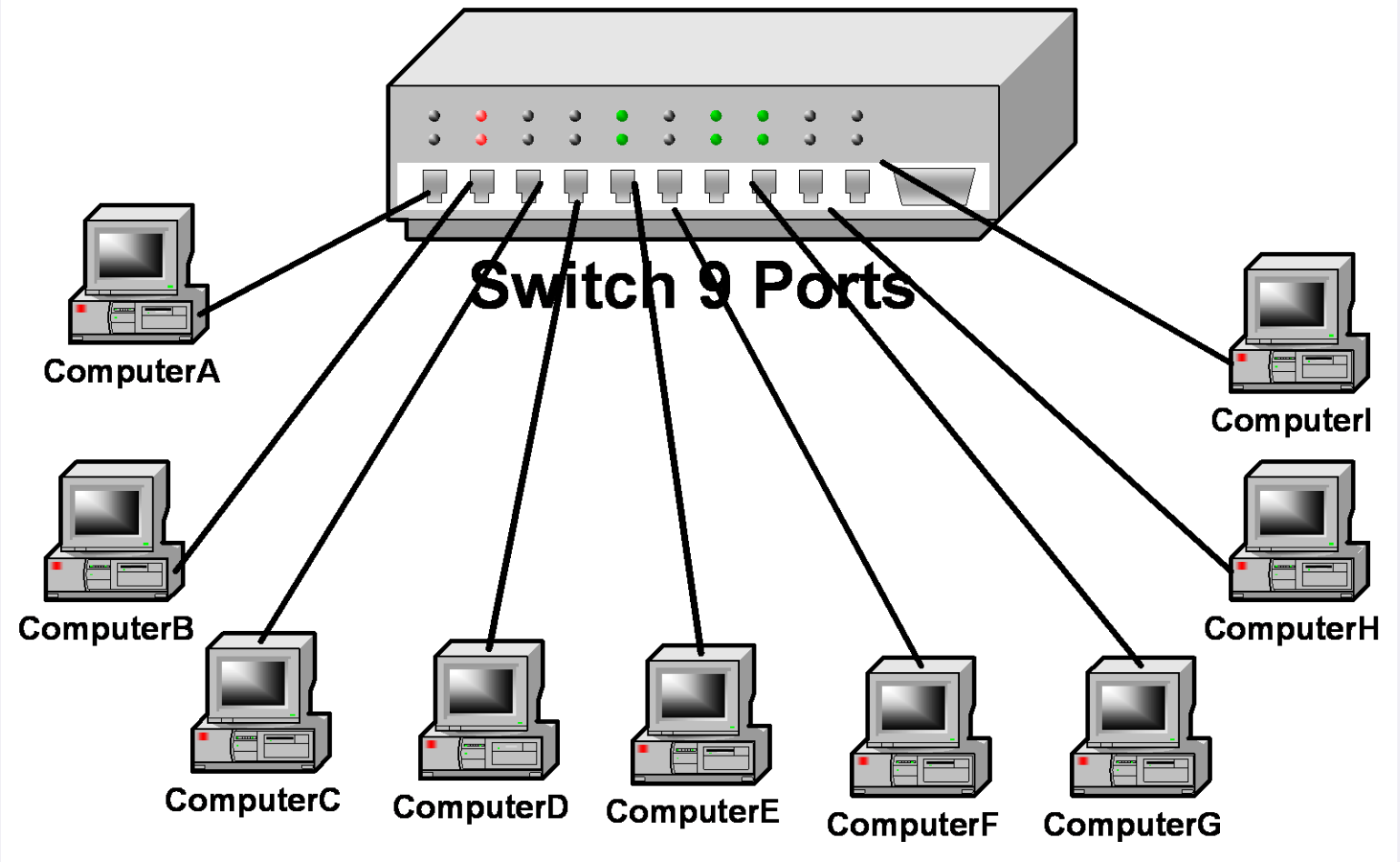
Segmenting with Switches (Micro segmentation)



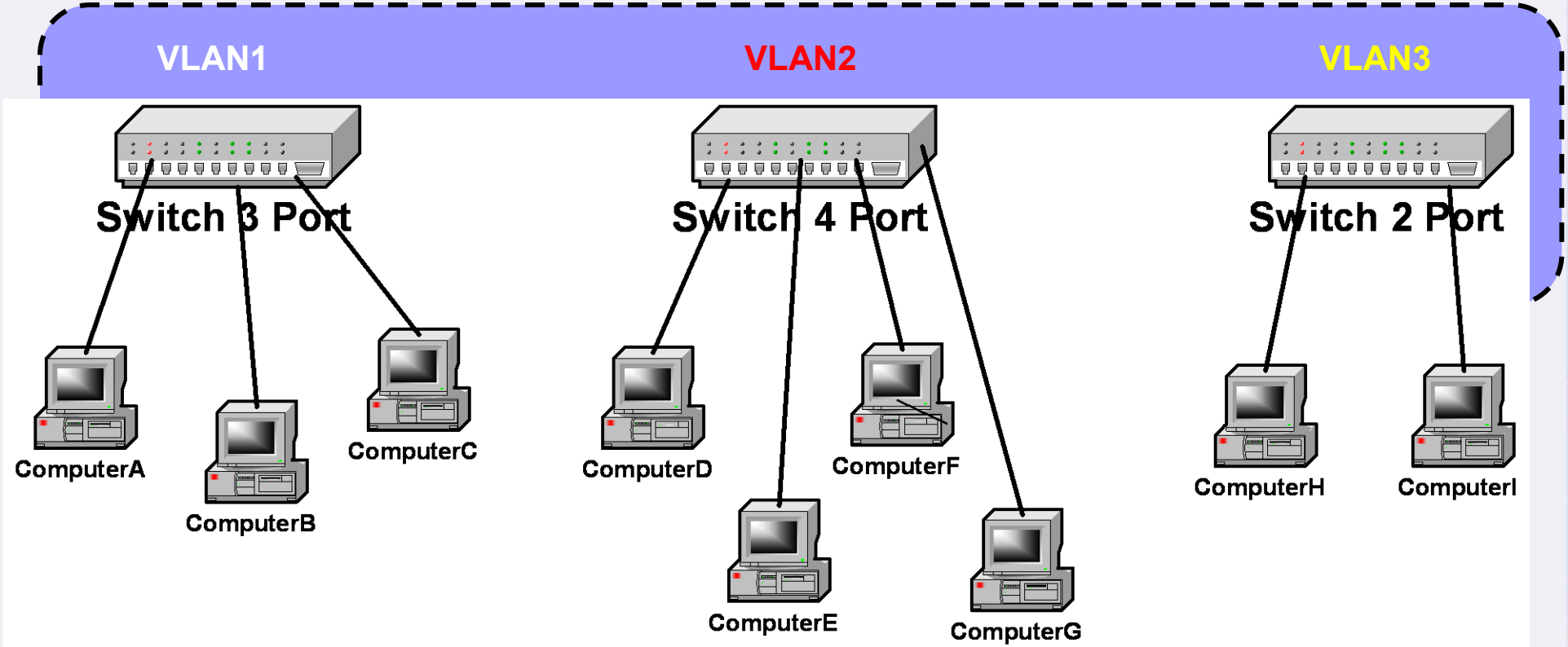
- A switch eliminates the impact of collisions through microsegmentation
- Low latencey and high frame-forwarding rates at each interface port
- Works with existing 802.3(CSMA/CD) compliant network interface cards and cabling

Virtual Local Area Network (VLAN)

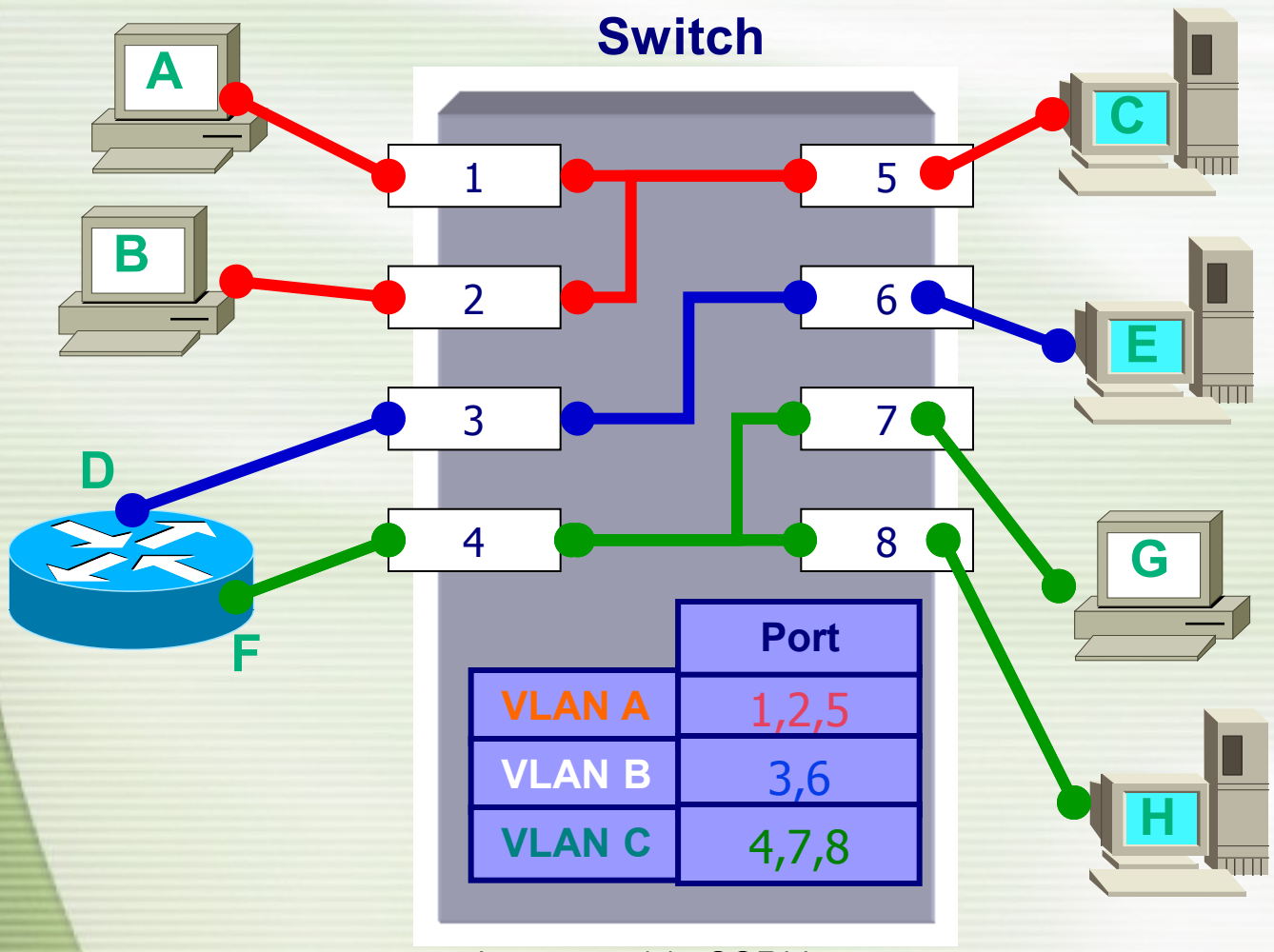
What is Virtual LAN



What is Virtual LAN ? (Cont.)



Virtual LAN (VLAN)

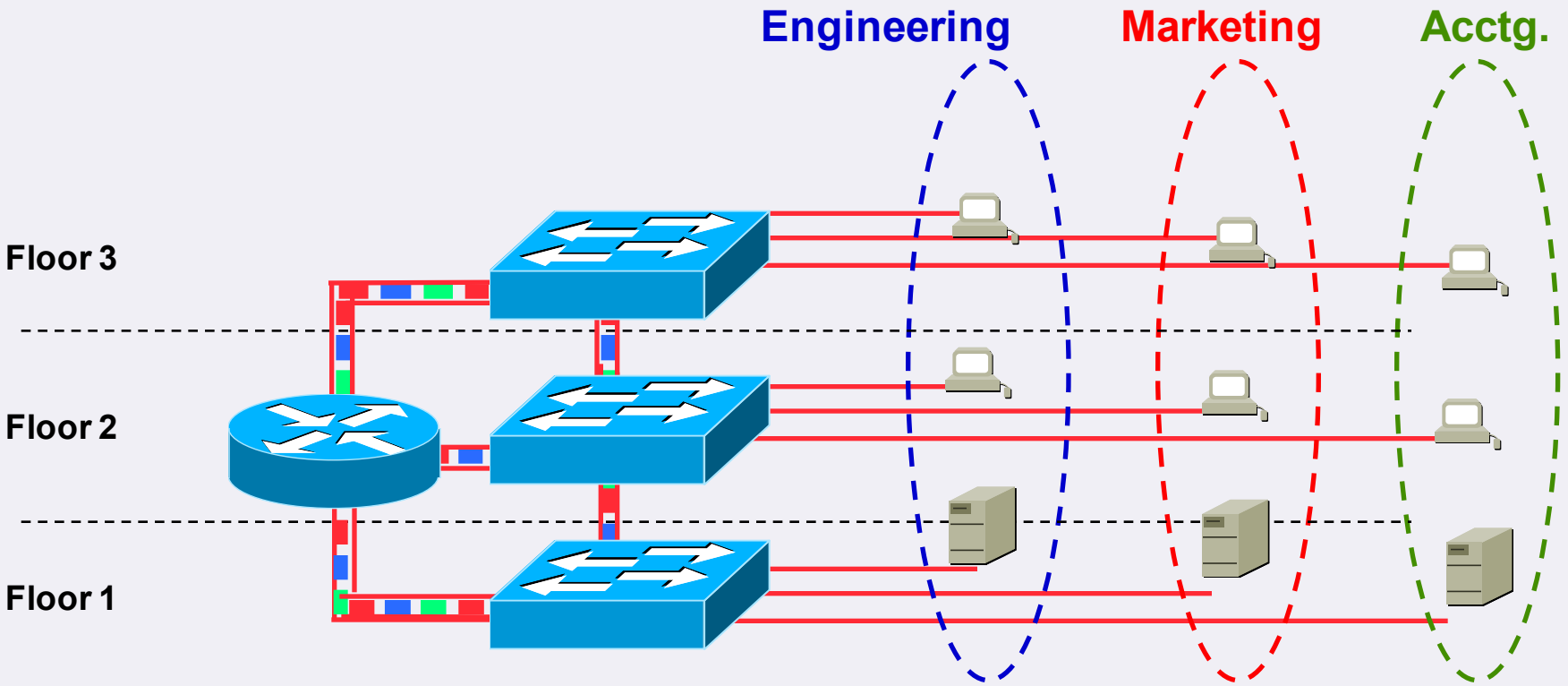


VLAN A	
Add.	Port
A	1
B	2
C	5

VLAN B	
Add.	Port
D	3
E	6

VLAN C	
Add.	Port
F	4
G	7
H	8

Remove the Physical Boundaries

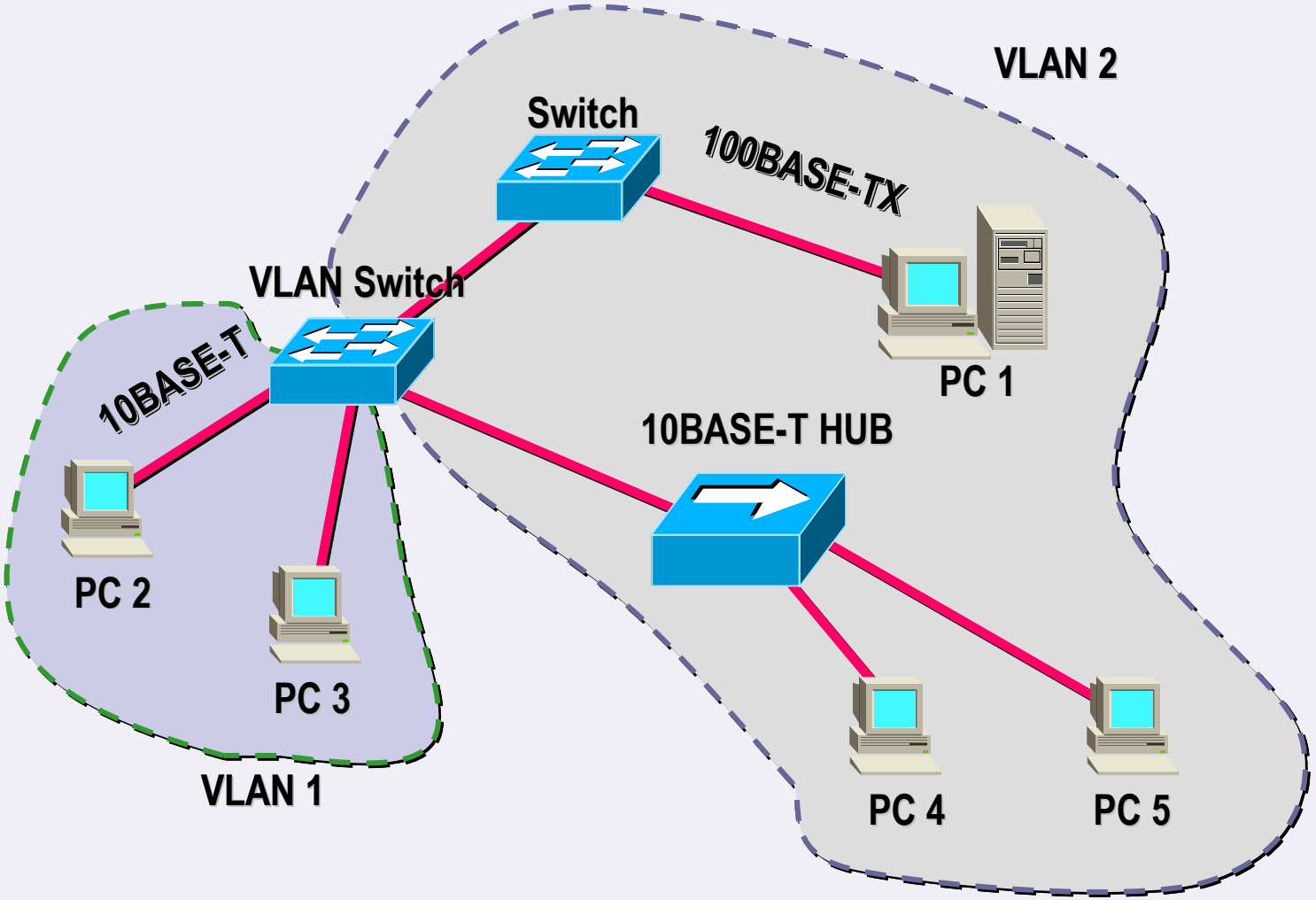


- Group users by department, team, or application
- Routers provide communication between VLANs

VLAN Benefits

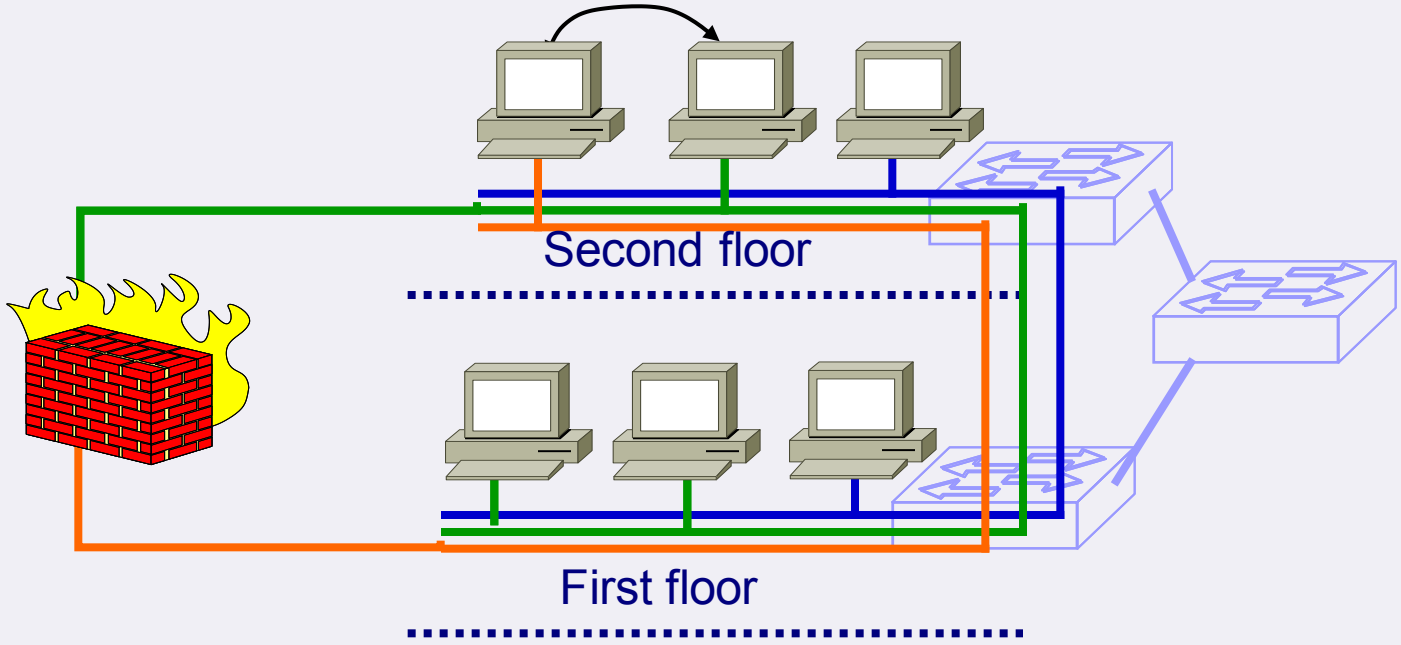
- Reduced administrative costs
  - Simplify moves, adds, and changes
- Efficient bandwidth utilization
  - Better control of broadcasts
- Improved network security
  - Separate VLAN group for high-security users
  - Relocate servers into secured locations
- Scalability and performance
  - Microsegment with scalability
  - Distribute traffic load

Better Control of Broadcasts

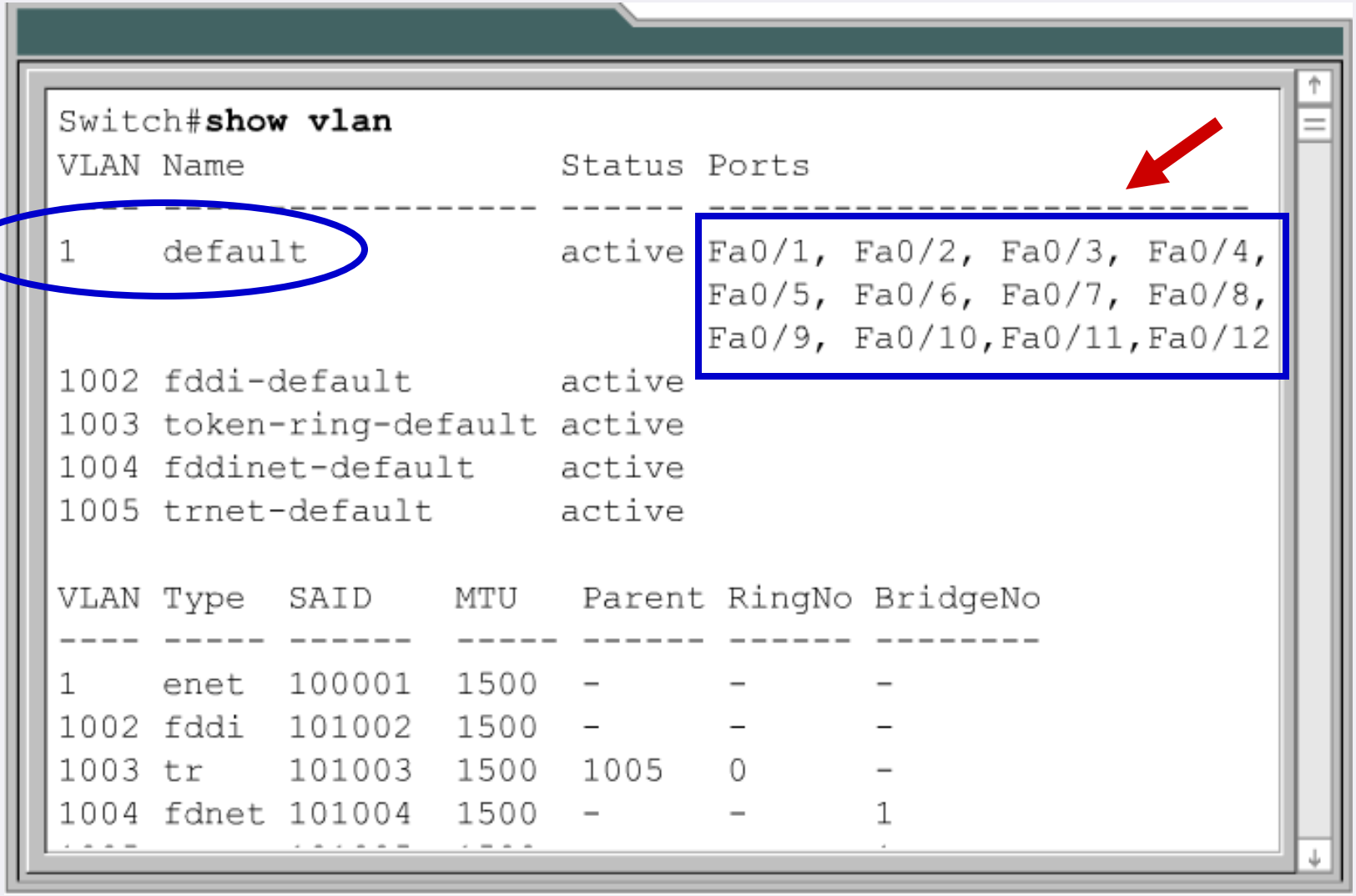


Separate VLAN Group for High-Security Users

- High-security can be grouped into a VLAN, no users outside of that VLAN can communicate with them.
- Inter VLANs communication must be achieved via routers (Layer 3) which are more filtering capability, i.e., more secure.



Default VLAN Configuration of Catalyst Switch



```
Switch#show vlan
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4,
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8,
                                                Fa0/9, Fa0/10,Fa0/11,Fa0/12
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo
---- ----- ---------- ----- ------ ------ --------
1    enet  100001     1500  -      -      -
1002 fddi  101002     1500  -      -      -
1003 tr    101003     1500  1005   0      -
1004 fdnet 101004     1500  -      -      1
```

Applicable Assigning the Port Mode and VLAN ID

Cisco

```
Enter configuration commands, one per line. End with CNTL/Z.

Switch#config terminal
Switch(config)#interface fastethernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#
```





Verifying VLAN Configuration (“show vlan” Command)

Cisco

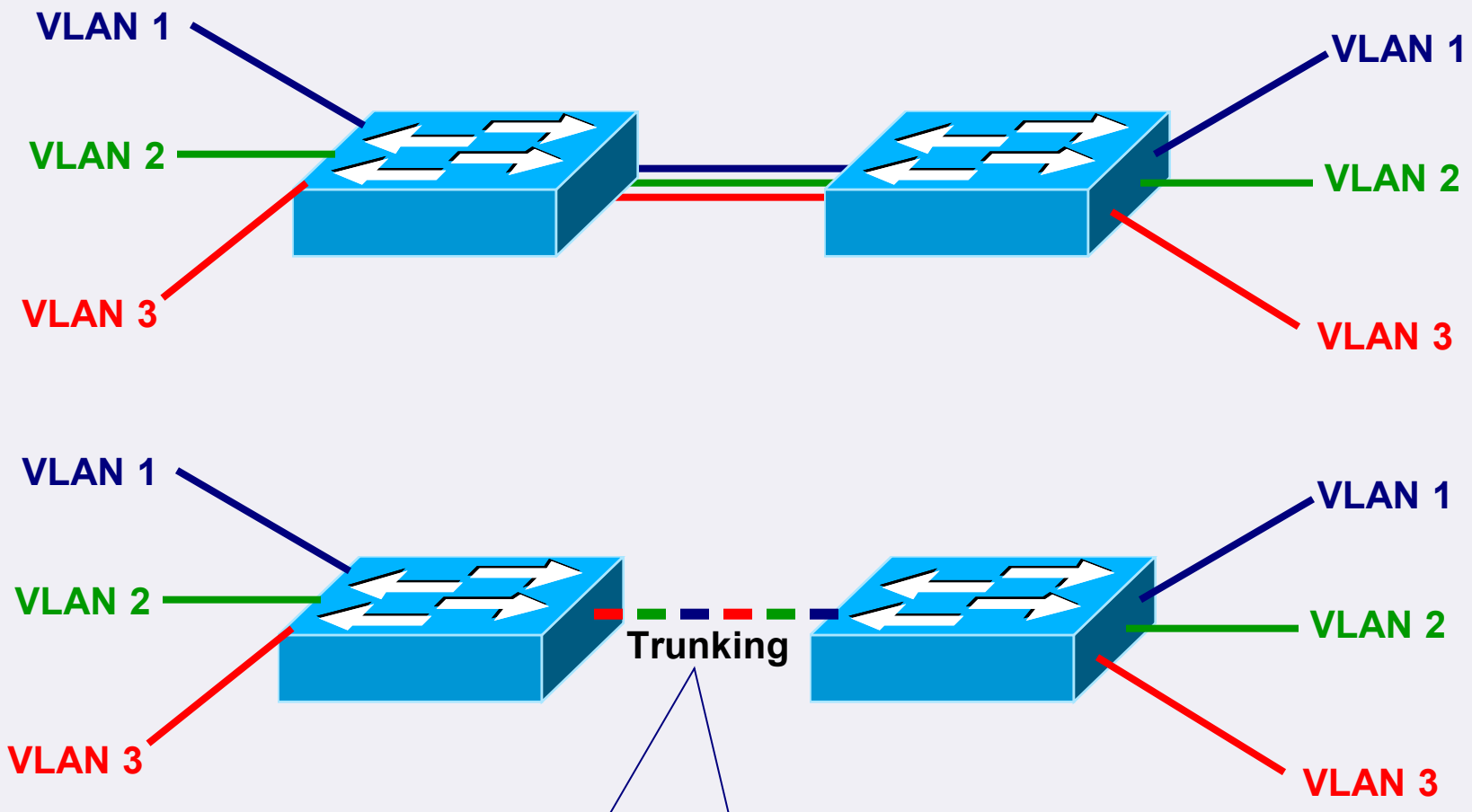
```
SydneySwitch#show vlan

VLAN Name                 Status    Ports
---- -------------------- ------- -------------------------------

VLAN Name                 Status    Ports
---- -------------------- ------- -------------------------------
1    default              active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
2    VLAN2                active    Fa0/5, Fa0/6, Fa0/7
3    VLAN3                active    Fa0/8, Fa0/9, Fa0/10, Fa0/11,
                                    Fa0/12
1002 fddi-default         active
1003 token-ring-default   active
1004 fddinet-default      active
1005 trnet-default        active

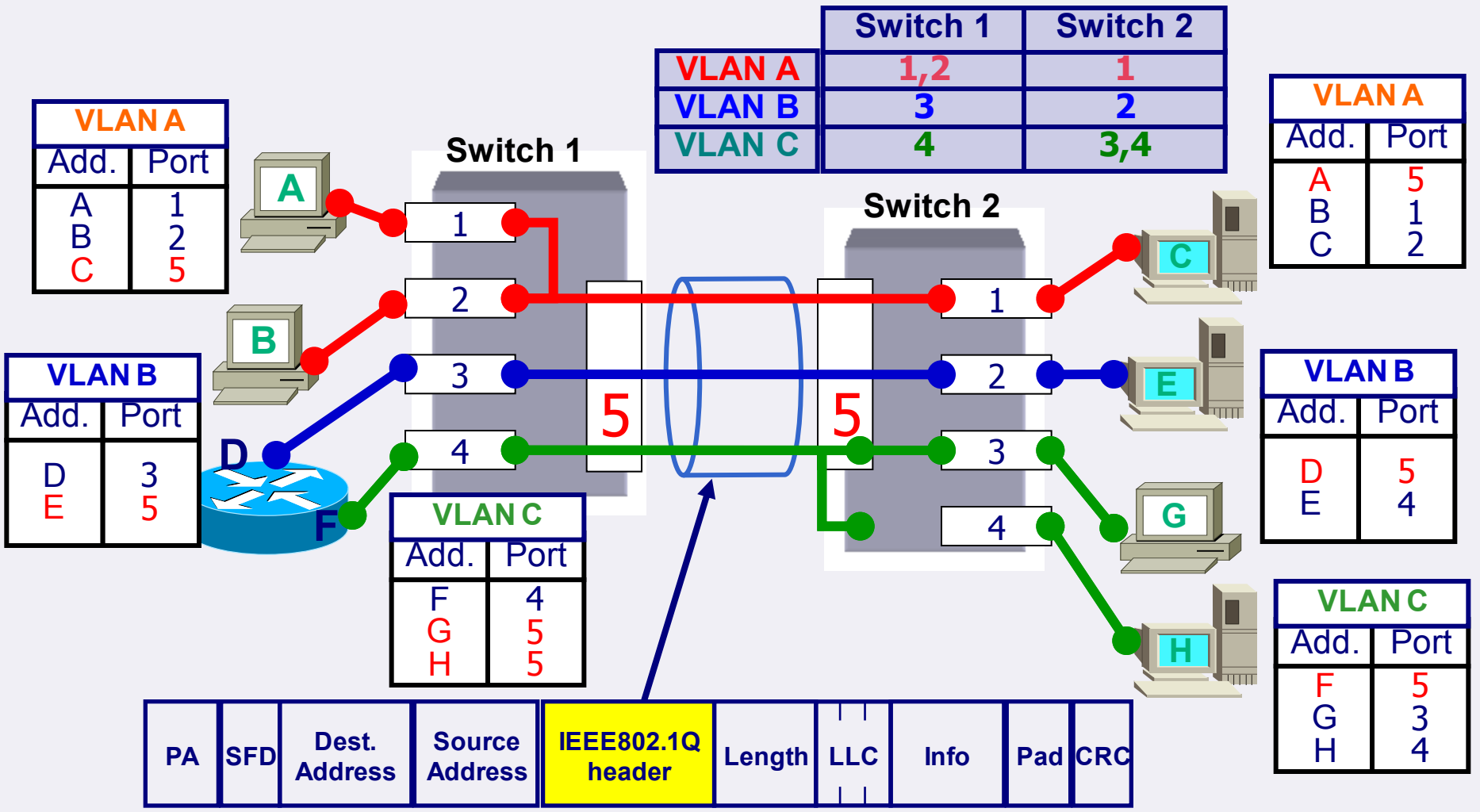
VLAN Type SAID MTU  Parent RingNo BridgeNo Stp BrdgMode Trans1 Trans2
---- ---- ---- ---- ------ ------ -------- ---------- ------ -------
1    enet 100001 1500  -      -      -        -    -      1002   1003
2    enet 100002 1500  -      -      -        -    -      0      0
```

VLAN Trunking

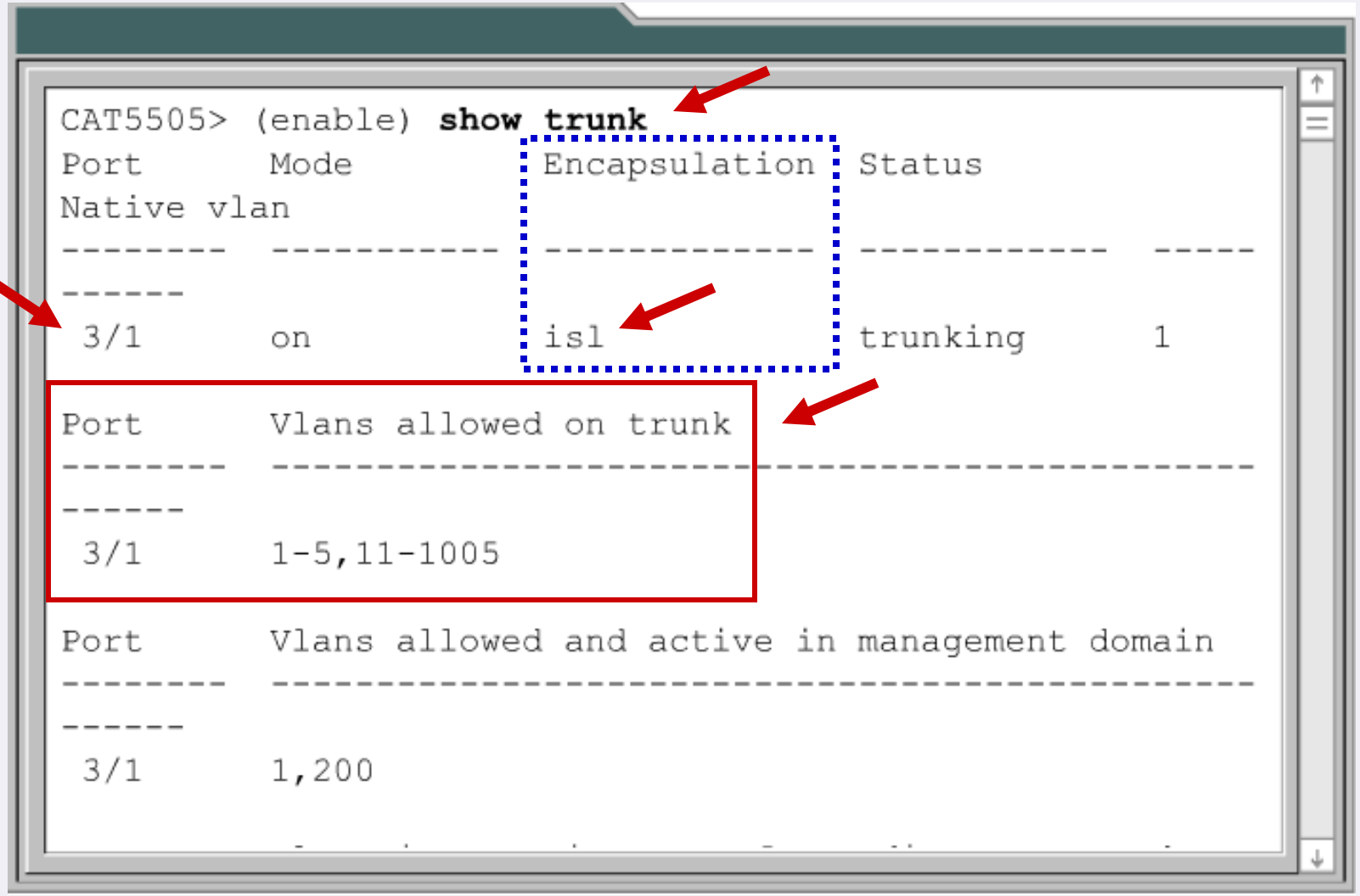


- **ISL (Inter – Switch Link)**
- **IEEE 802.1q**

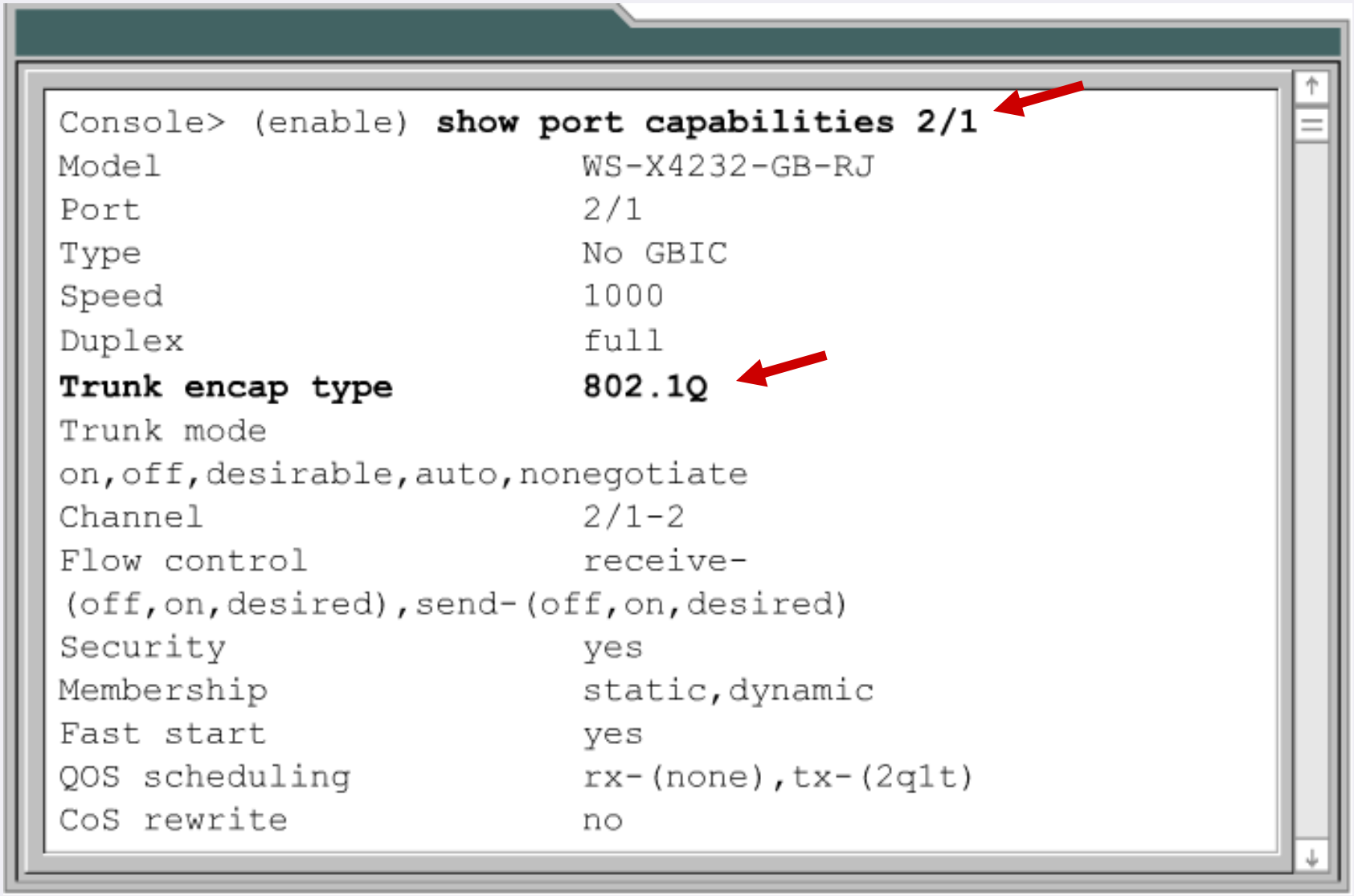
IEEE 802.1Q VLAN Tagging Scheme



Verifying the Trunk Configuration



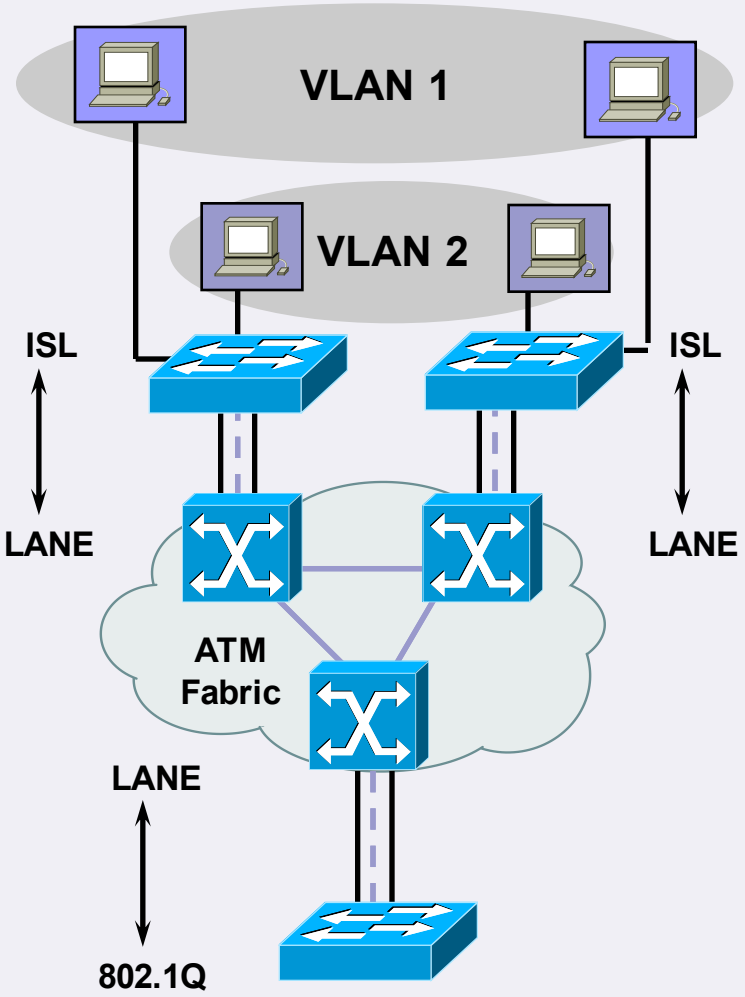
The “show port capabilities” Command



```
Console> (enable) show port capabilities 2/1
Model                     WS-X4232-GB-RJ
Port                      2/1
Type                      No GBIC
Speed                     1000
Duplex                    full
Trunk encap type          802.1Q
Trunk mode
on,off,desirable,auto,nonegotiate
Channel                   2/1-2
Flow control              receive-
(off,on,desired),send-(off,on,desired)
Security                  yes
Membership                static,dynamic
Fast start                yes
QOS scheduling            rx-(none),tx-(2q1t)
CoS rewrite               no
```

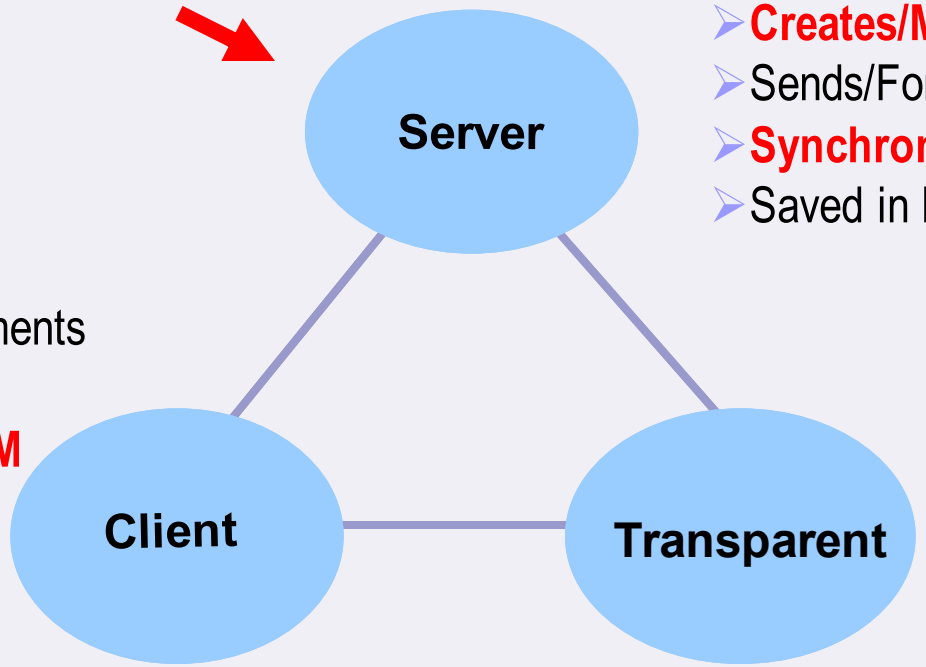
VLAN Trunk Protocol (VTP) “Cisco Proprietary”

- VLAN administration and configuration protocol
  - Reduces VLAN setup and administration
  - Eliminates configuration errors
  - Maps VLANs across different backbones (FDDI, Fast Ethernet, ATM)
  - Maps between ISL and 802.1q
  - Maintains security between VLANs



VTP Modes

Default VTP Mode



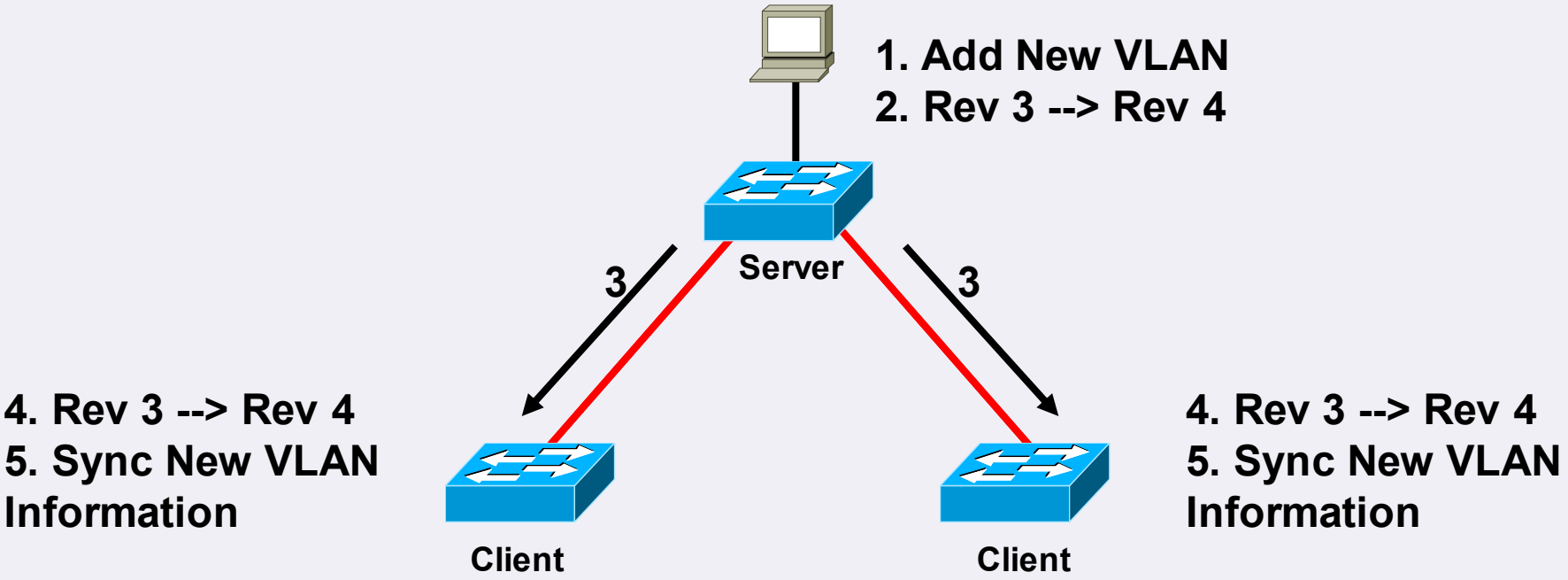
- Creates/Modifies/Deletes VLANs
- Sends/Forwards Advertisements
- Synchronizes
- Saved in NVRAM

- Forwards Advertisements
- Synchronizes
- Not saved in NVRAM

- Creates/Modifies/Deletes VLANs
- Forwards Advertisements
- Does not Synchronize
- Saved in NVRAM

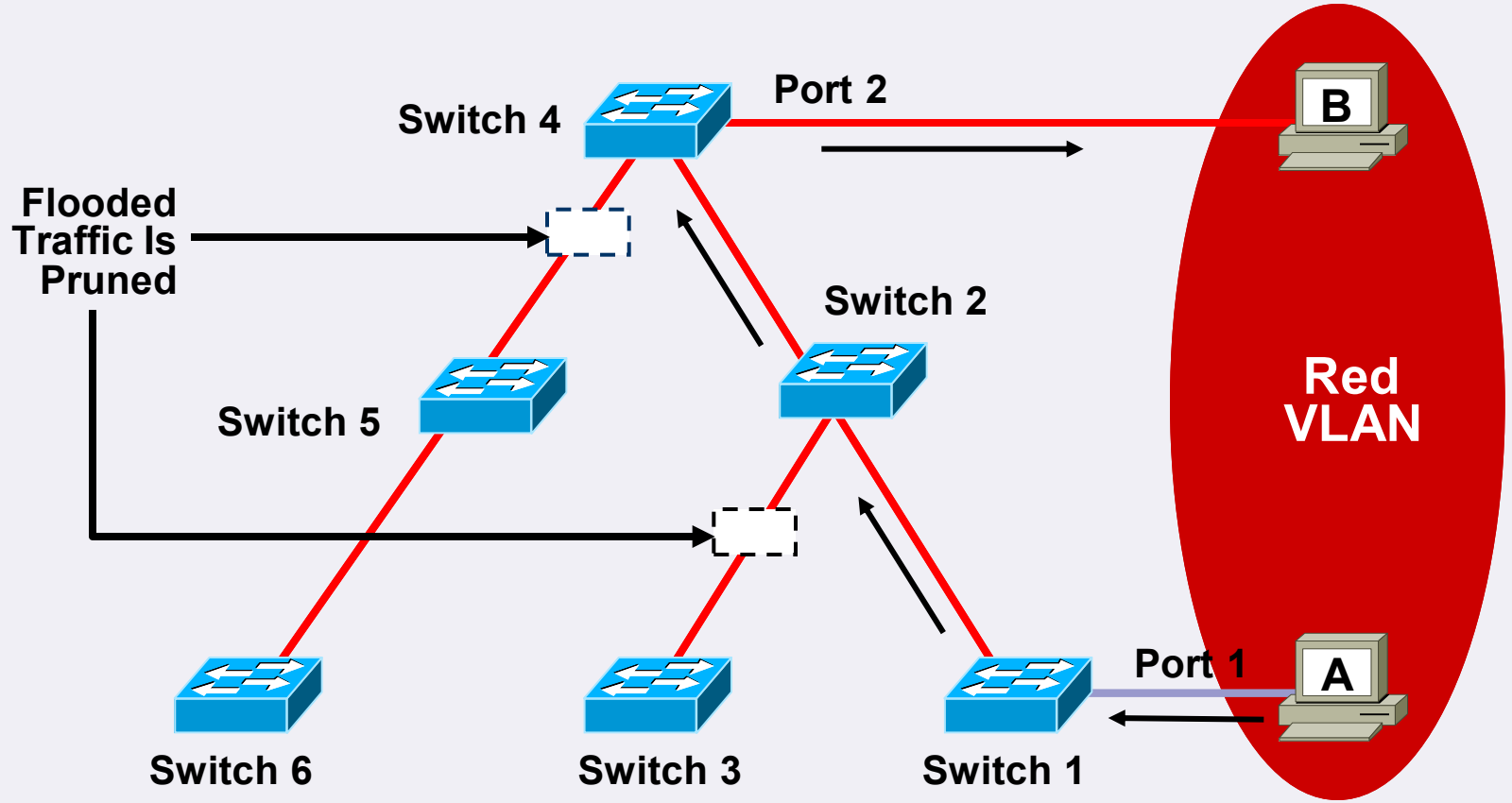
How VTP Works

- VTP advertisements are sent as multicast frames.
- VTP servers and clients are synchronized to the latest revision number.
- VTP advertisement are sent every 5 minutes or when there is a change.



VTP Pruning

- Increases available bandwidth by reducing unnecessary flooded traffic
- Example: Station A sends broadcast, and broadcast is flooded only toward any switch with ports assigned to the red VLAN



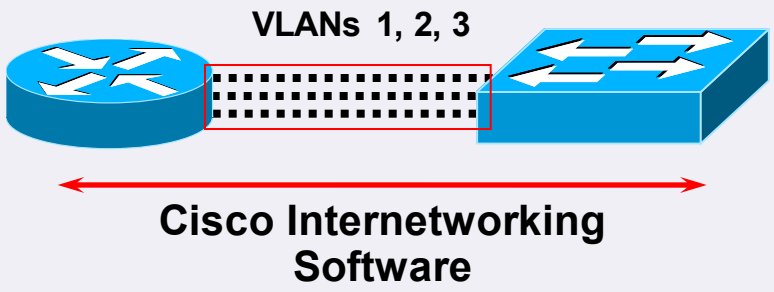
Checking the VTP Status

```
MDF Switch#show vtp status
VTP Version                          :2
Configuration Revision               :0
Maximum VLANs supported locally      :64
Number of existing VLANs             :7
VTP Operation Mode                   :Server
VTP domain Name                      :cisco
VTP Pruning Mode                     :Disabled
VTP V2 Mode                          :Enabled
VTP Traps Generation                 :Disabled
MDS digest                           :0x30 0x50
Configuration last modified by 10.1.1.252 a local
updater ID 138.25.13.121 on interface found)
MDF Switch#exit
```

Communicating Between VLANs

Two Physical Topology Approaches

Logical Communication

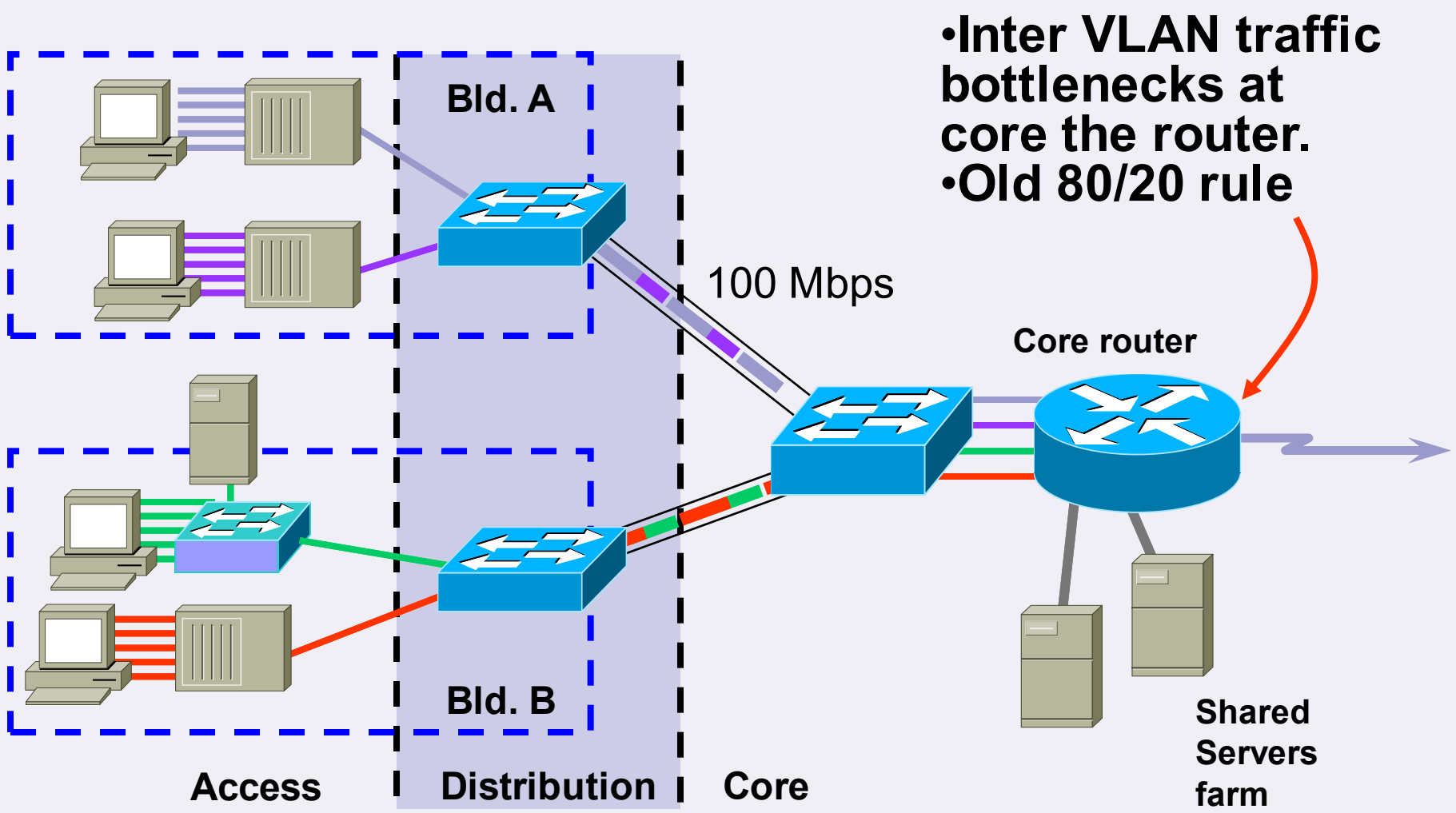


Physical Link per VLAN



- Layer 3 links VLANs together
- Adds additional security and management
- Logical links conserve physical ports
- Multimode, depending on protocol
- Controls access by VLAN
- Up to 255 VLANs per router

Switched LAN: VLAN with Centralized Routing



Spanning Tree Protocol Overview

Spanning Tree Process

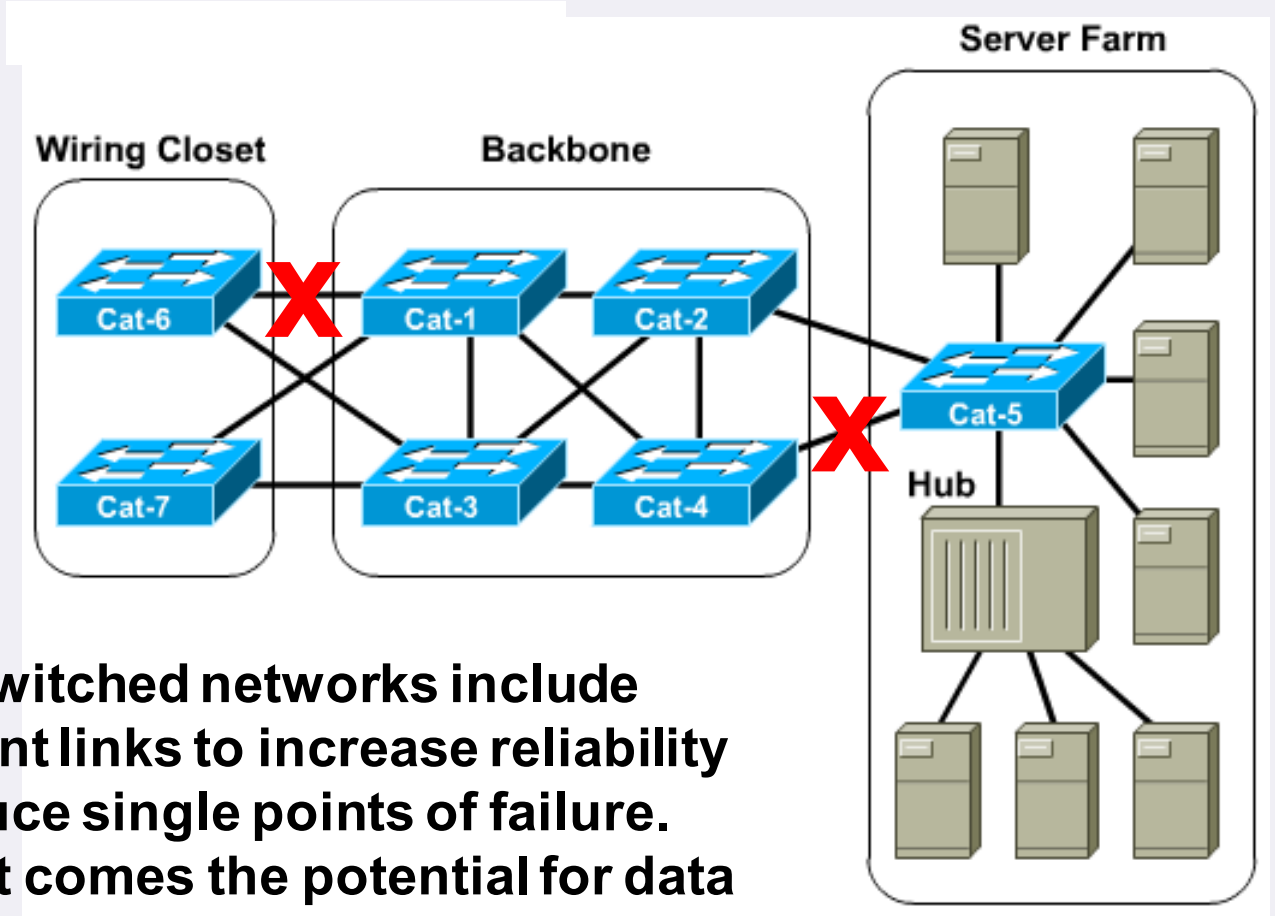
- **802.1D** is the IEEE specification for STP.
- STP facilitates secondary paths to a destination.
- Its main purpose is however to remove **switching loops from networks with redundant switched paths** (i.e. layer 2 networks consisting of switches and bridges) and **broadcast storms** as each switch broadcasts data to every other switch.
- It constantly monitors the switch ports and makes sure the protocol knows about all of the links in your network. If more than one link exists, STP disables the secondary link until it is needed. This way the switch shuts down redundant links, putting a stop to any data loops present in the network.

Spanning Tree Process ....Cont.

- STP elects a root bridge in the network that will decide on the network topology by calculating a bridge ID (each network can only have one given root bridge).
- Other switches on your network are called non-root bridges.
- **Spanning Tree Algorithm (STA)** is used to determine the cost to assign to each link based on the bandwidth of each hop from one switch in the network to another switch. Manual costs can also be assigned to each port (**sum of the costs based on bandwidth**)
- STP is enabled on most Cisco switches with Ethernet or FastEthernet ports.

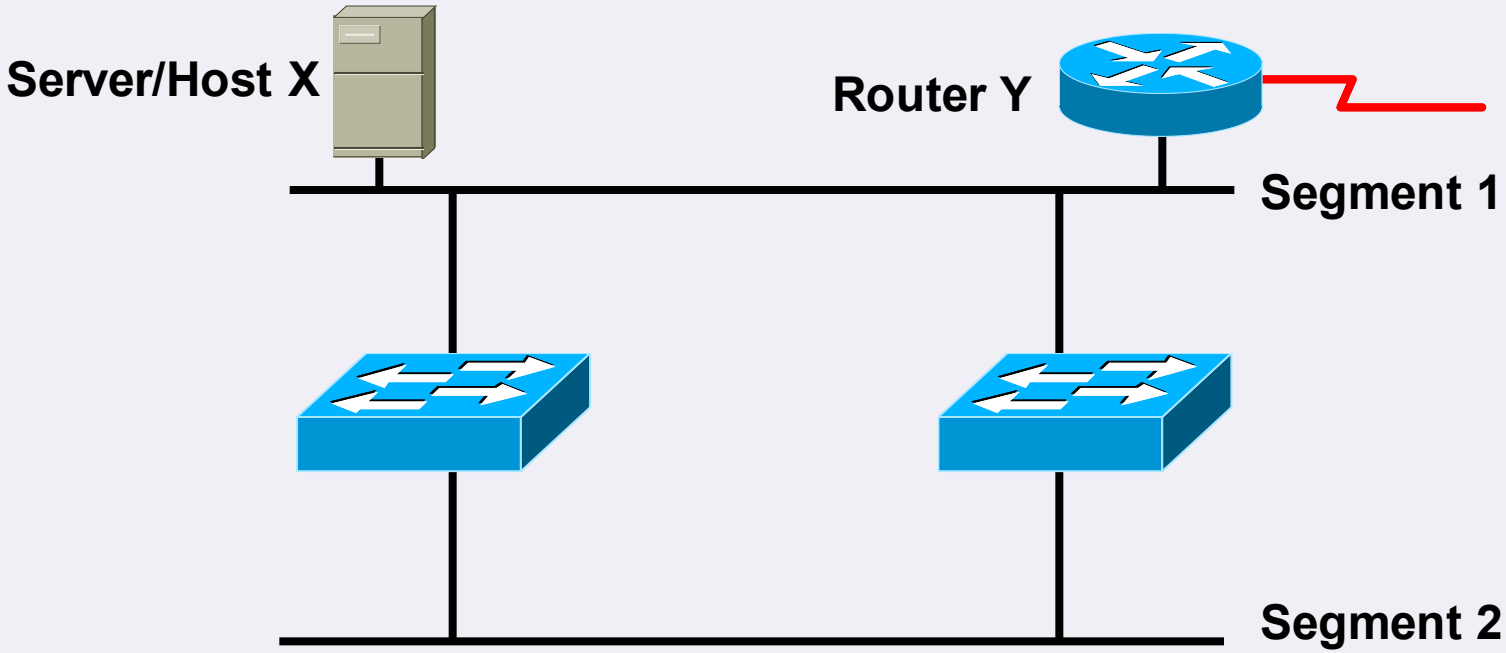
Ensuring Network Availability

Redundancy and Loops!!!



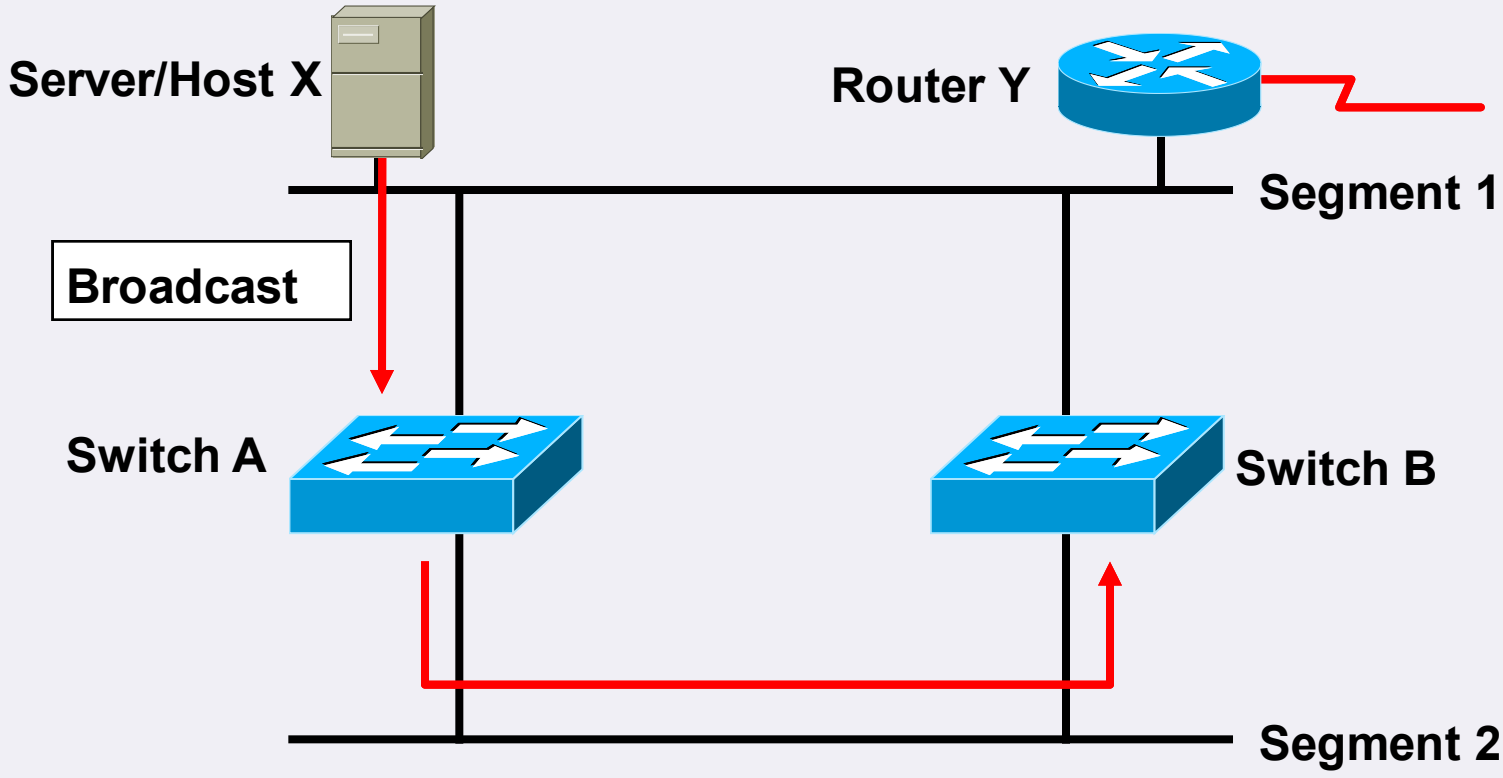
Newer switched networks include redundant links to increase reliability and reduce single points of failure. With that comes the potential for data loops.

Redundant Topology



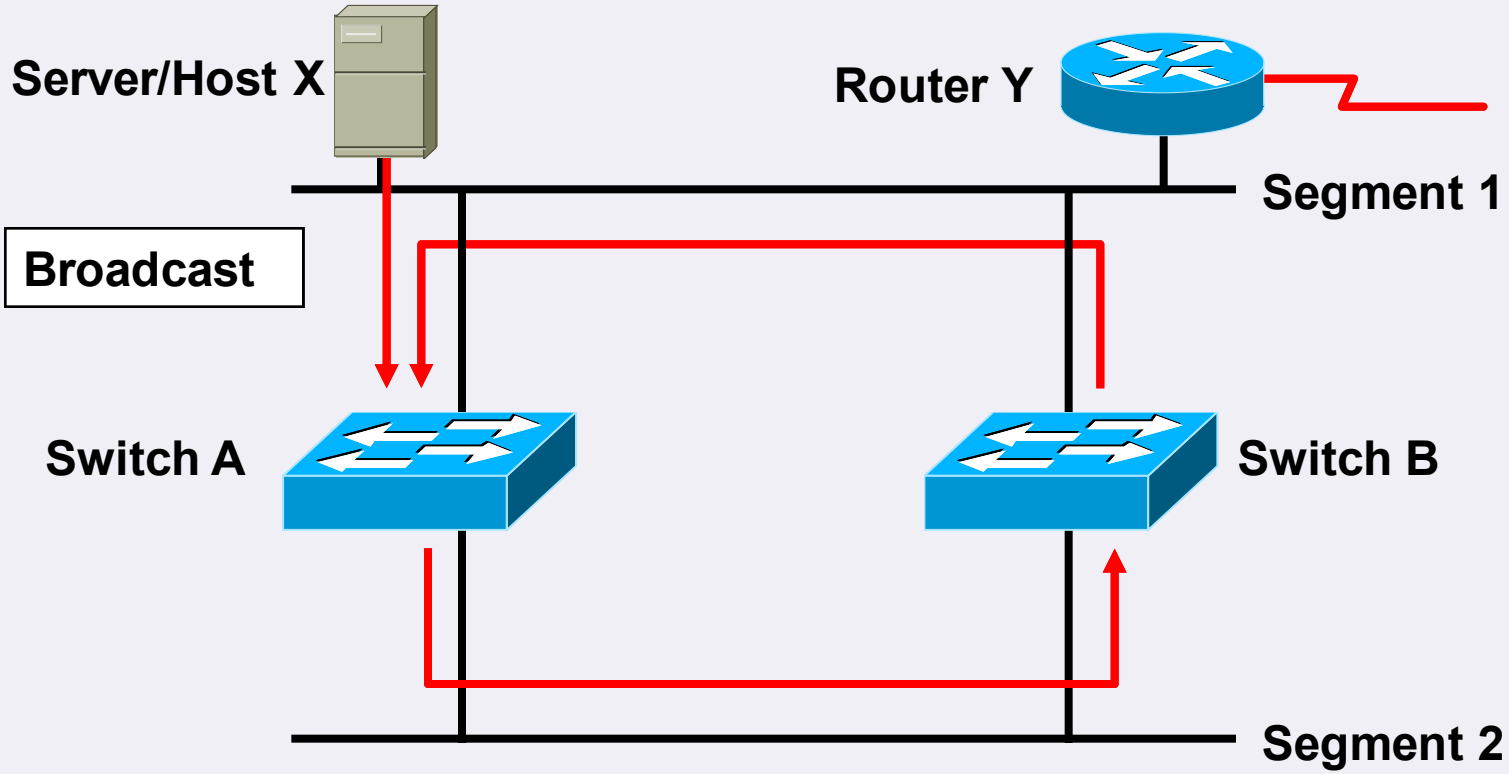
- Redundant topology eliminates single points of failure.
- Redundant topology causes broadcast storms, multiple frame copies, and MAC address table instability problems.

Broadcast Storms



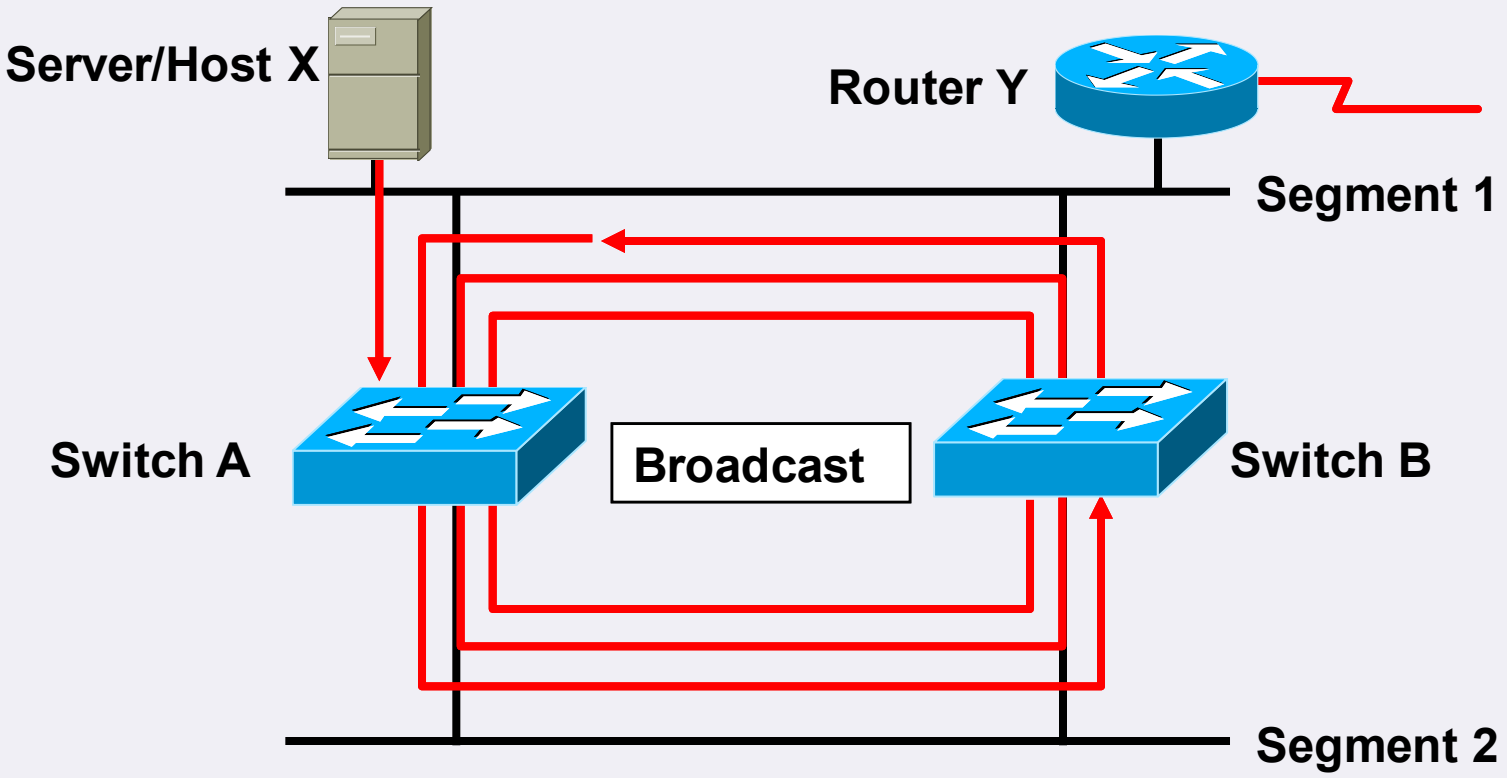
Host X sends a Broadcast

Broadcast Storms (cont.)



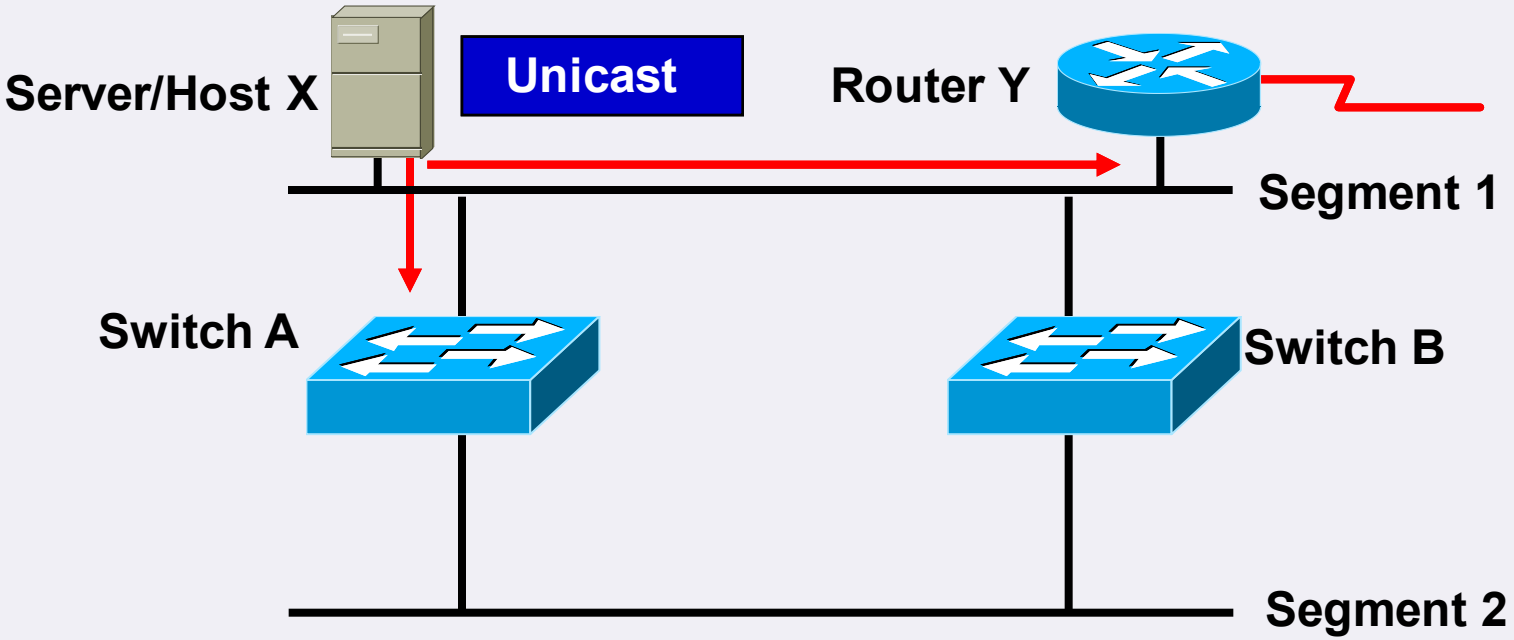
Host X sends a Broadcast

Broadcast Storms (cont.)



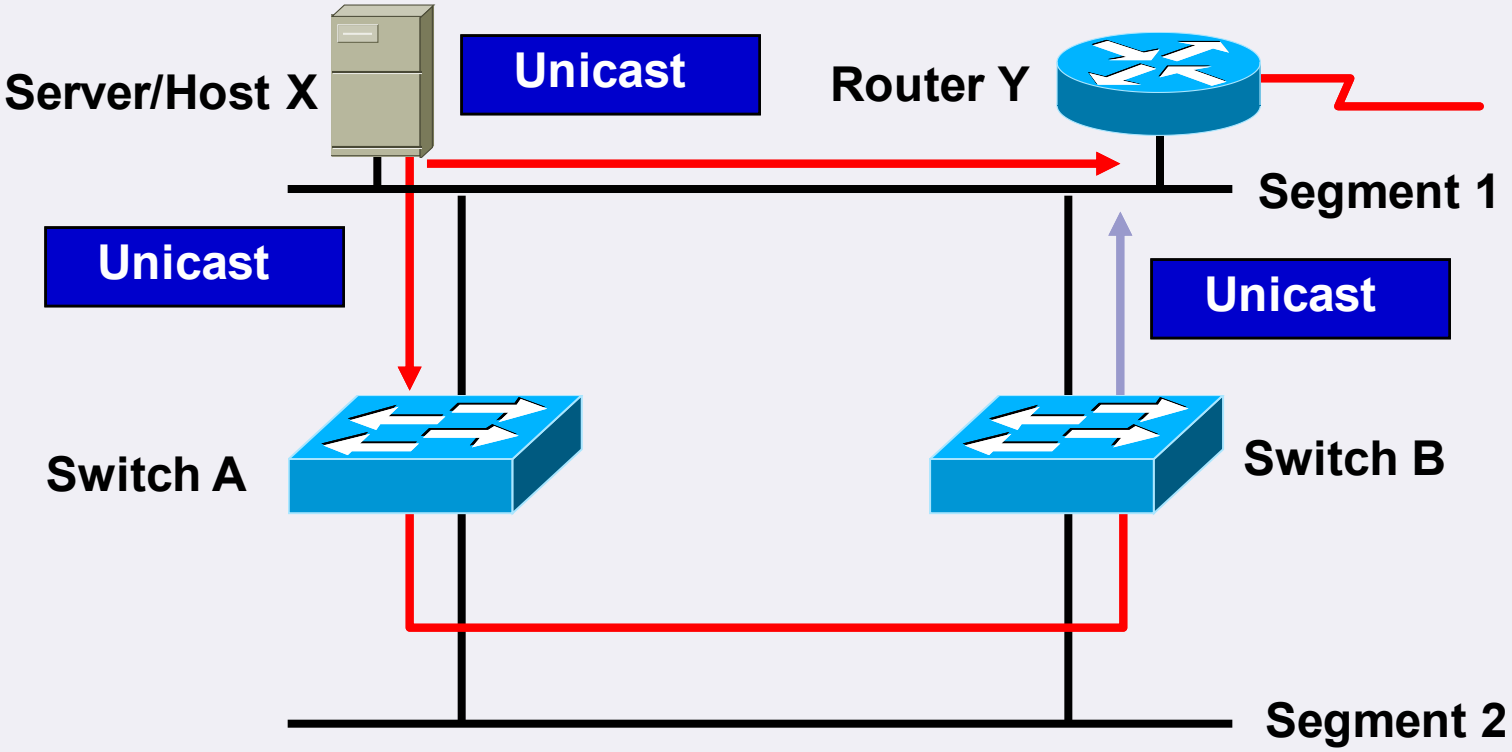
- **Switches continue to propagate broadcast traffic over and over.**

Multiple Frame Copies



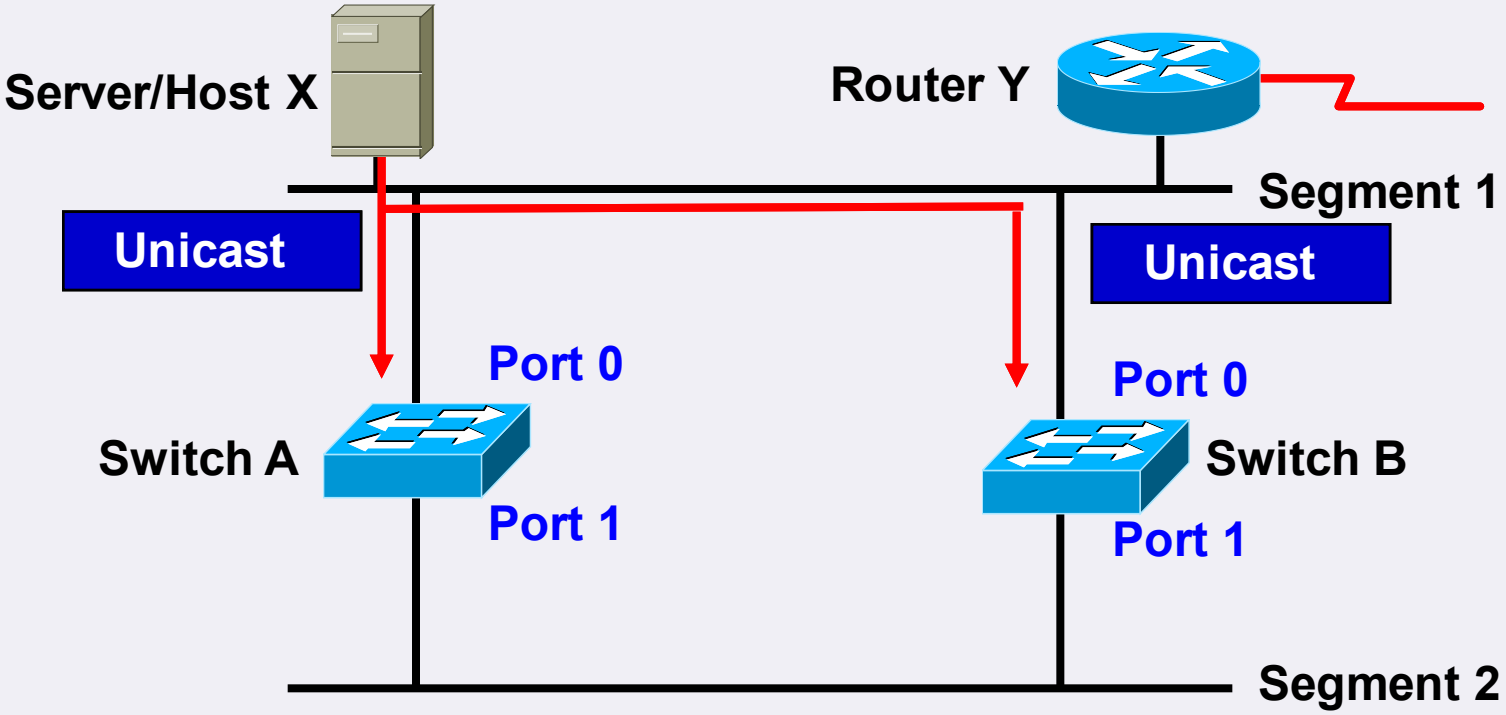
- Host X sends an unicast frame to router Y.
- Router Y MAC address has not been learned by either switch yet.

Multiple Frame Copies (cont.)



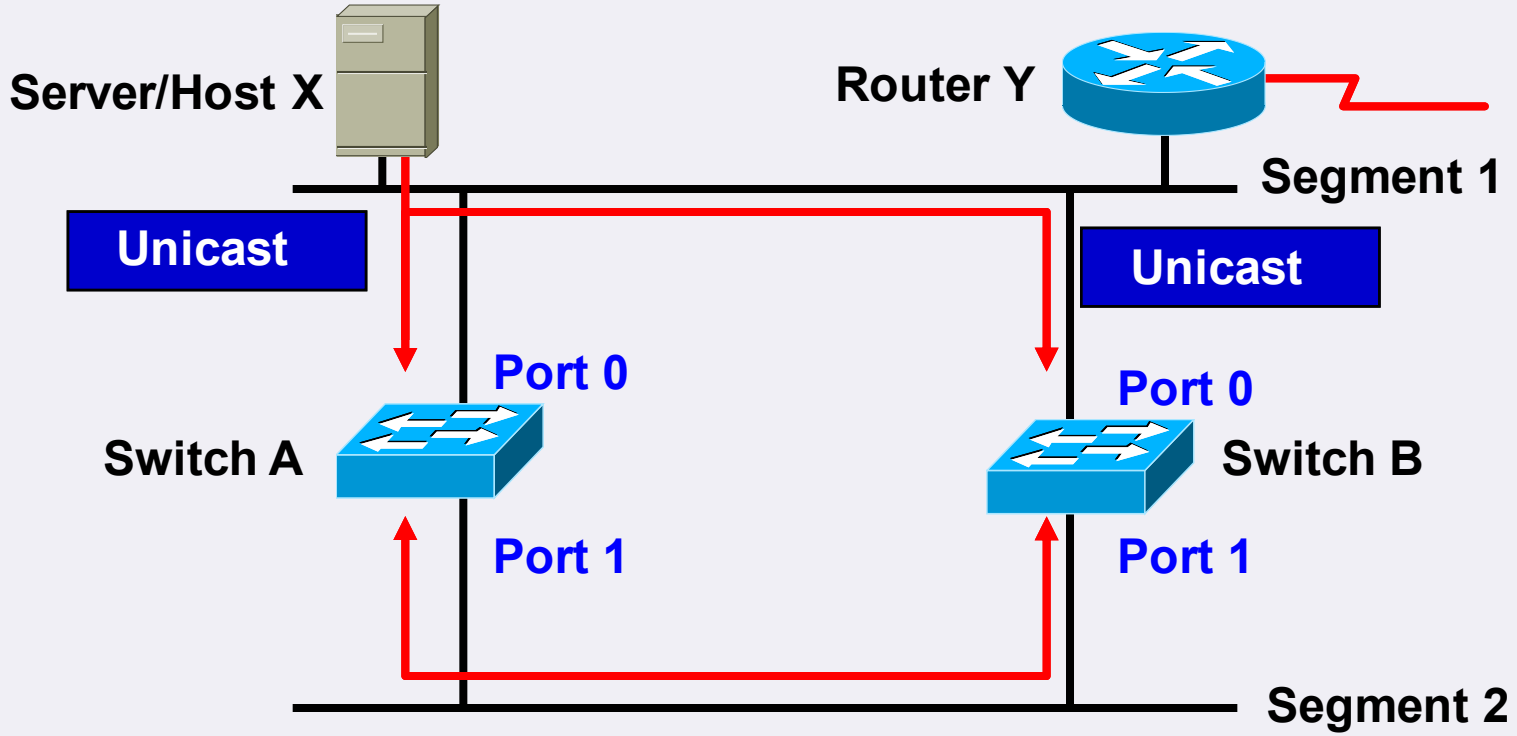
- Host X sends an unicast frame to router Y.
- Router Y MAC address has not been learned by either switch yet.
- Router Y will receive two copies of the same frame.

MAC Database Instability



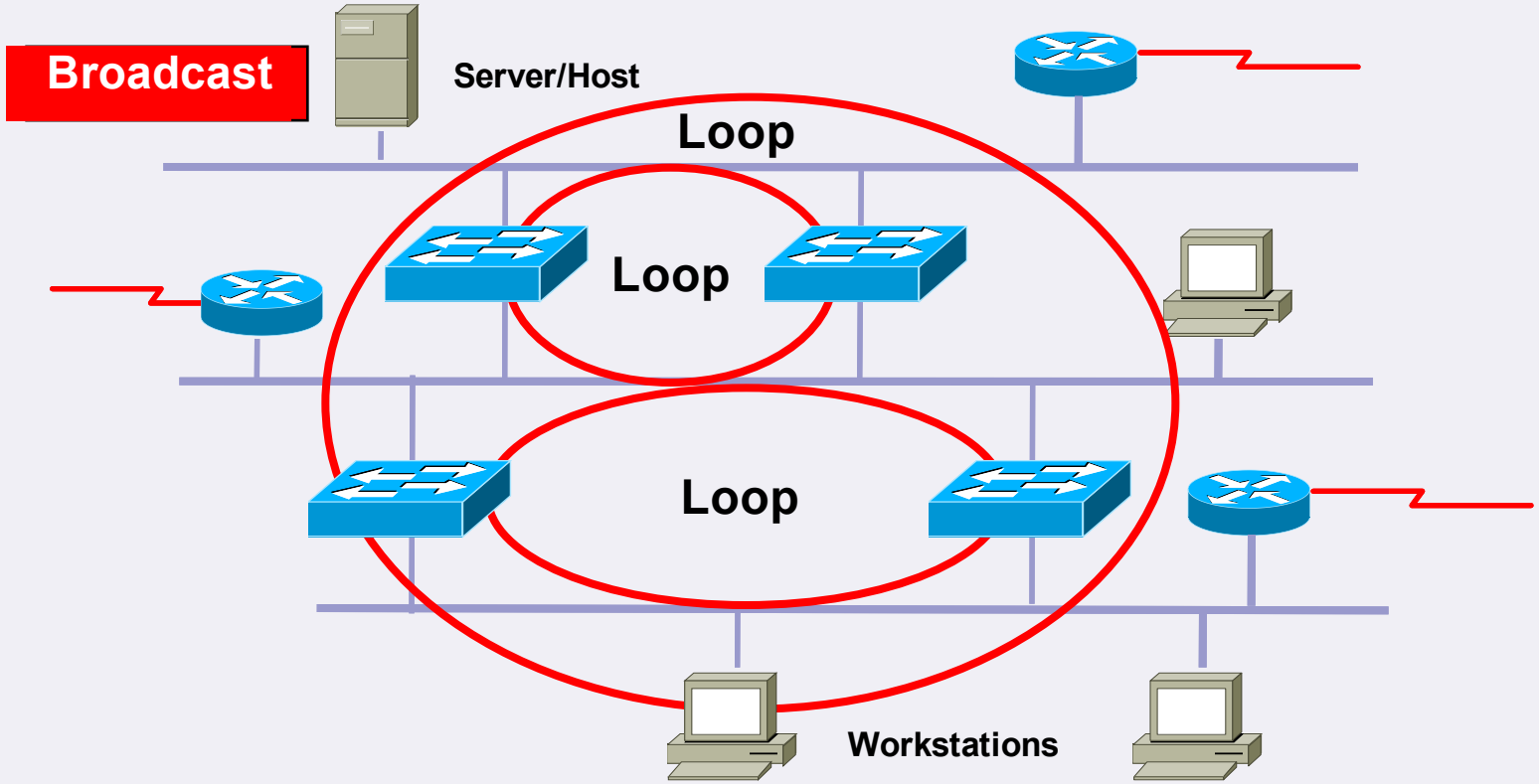
- Host X sends an unicast frame to router Y.
- Router Y MAC address has not been learned by either switch yet.
- Switch A and B learn the host X MAC address on port 0.

MAC Database Instability (cont.)



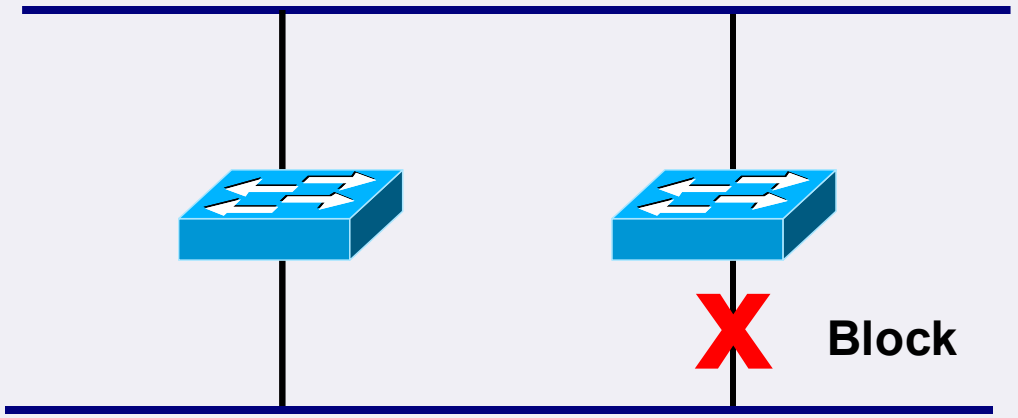
- Host X sends an unicast frame to router Y.
- Router Y MAC address has not been learned by either switch yet.
- Switch A and B learn the host X MAC address on port 0.
- The frame to router Y is flooded.
- Switches A and B incorrectly learn the host X MAC address on port 1.

Multiple Loop Problems



- Complex topology can cause multiple loops to occur.
- Layer 2 has no mechanism to stop the loop.

Solution: Spanning-Tree Protocol (802.1D)



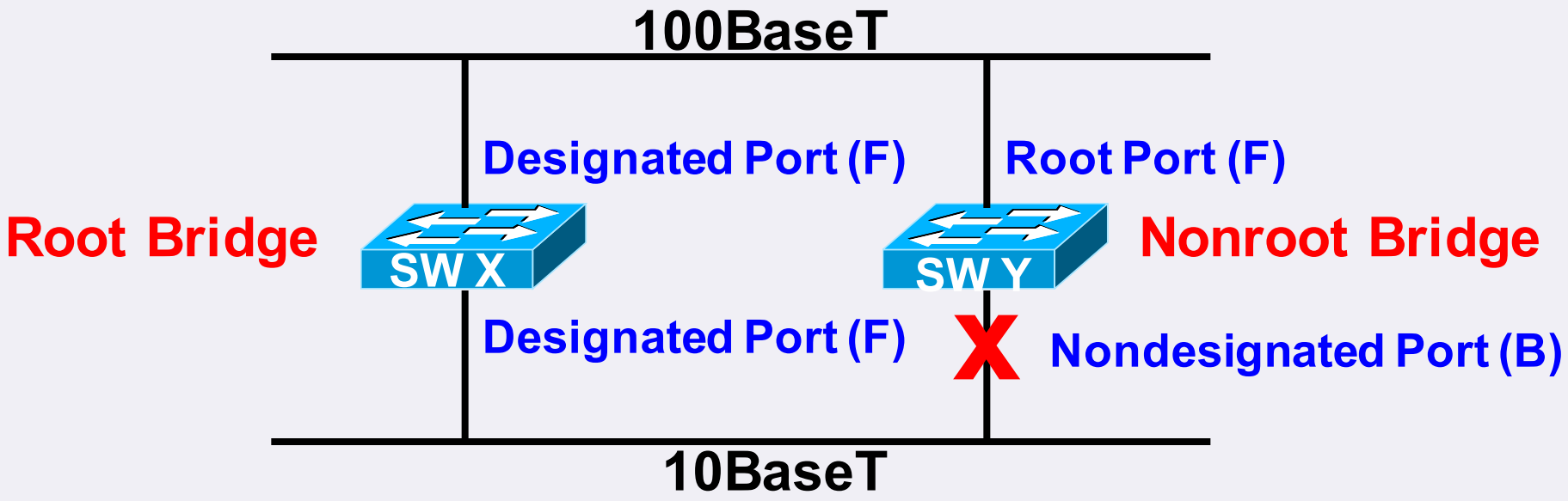
- Provides a loop-free redundant network topology by placing certain ports in the **blocking state**.

Spanning-Tree Operations

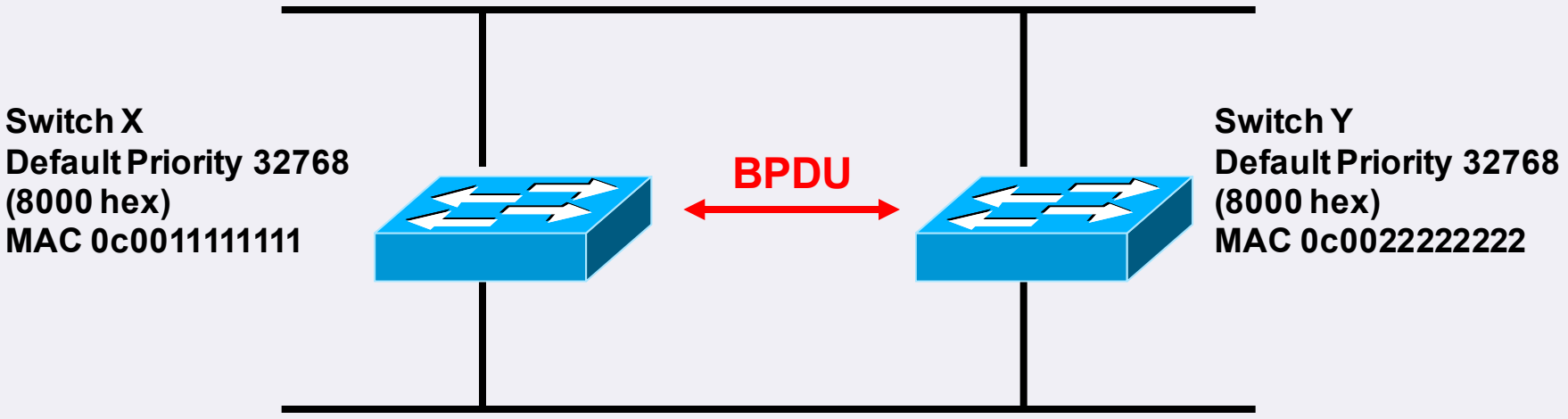
- **Selecting the Root Bridge**
  - Exchanging bridge information with Bridge Protocol Data Units (BPDU).
  - **Root Bridge** is the bridge with the lowest **bridge ID**.
    - ***Bridge ID = Bridge Priority (Default = 32768) + Bridge MAC address***
- **Selecting the Designated Port**
  - For the non-root bridge, there will be one **root port**.
    - *Root port is the lowest **root path cost**.*
  - On each segment, there will be one **designated bridge** & one **designated port**.
    - ***Designated bridge** is the bridge that has the lowest root path cost to the root bridge.*
    - ***Designated port** is the port on designated bridge that is selected to send or receive traffic from the segment.*
  - All other ports will become **blocked ports**.
- **Convergence**

Spanning-Tree Operations

- One root bridge per network
- One root port per nonroot bridge
- One designated port per segment

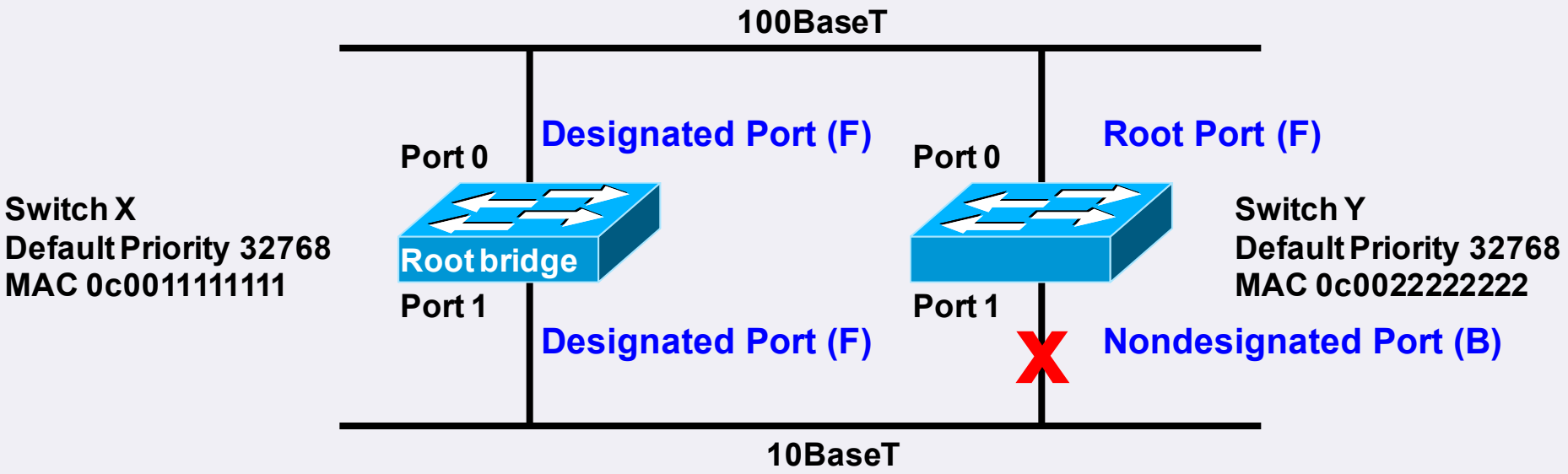


Spanning-Tree Protocol - Root Bridge Selection

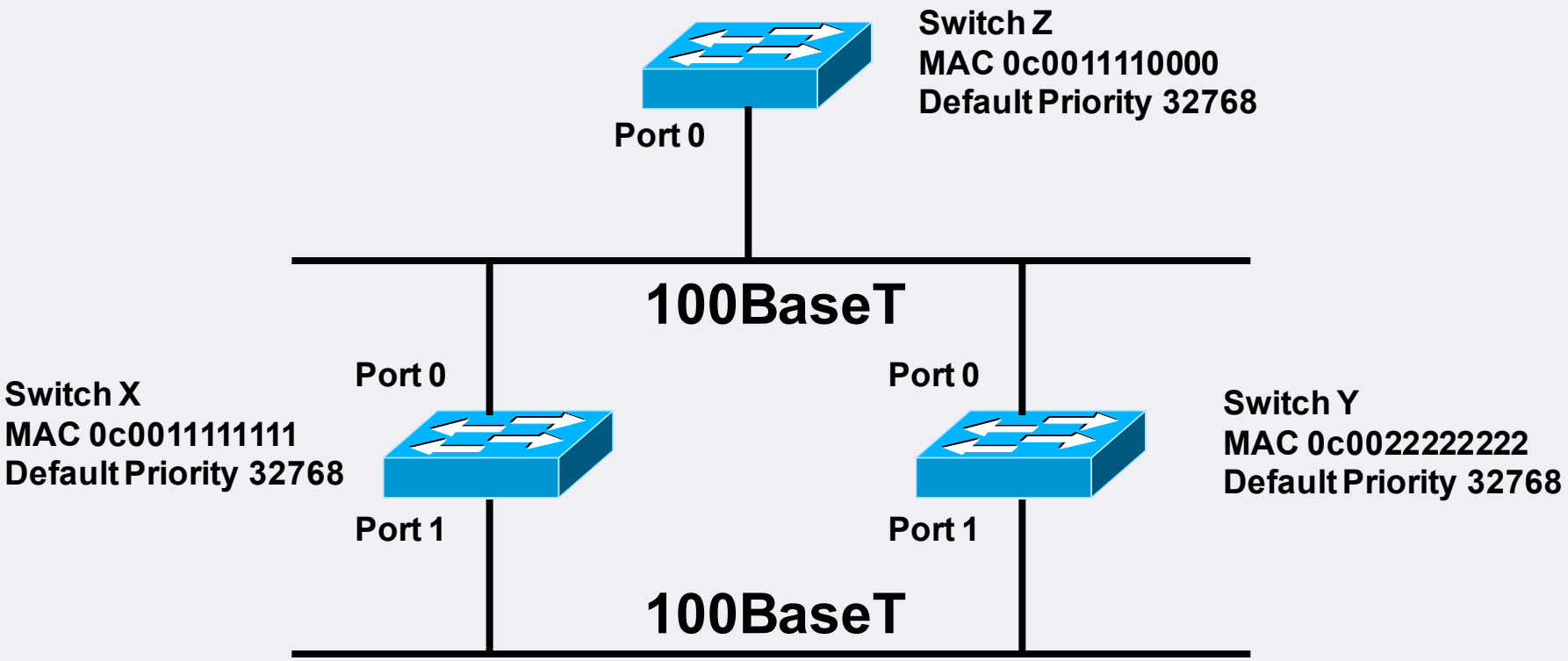


- BPDU = Bridge Protocol Data Unit
  (default = sent every 2 seconds).
- Root bridge = Bridge with the lowest bridge ID.
- Bridge ID = Bridge priority + bridge MAC address.
- In the example, which switch has the lowest bridge ID?

Spanning-Tree Protocol Port States

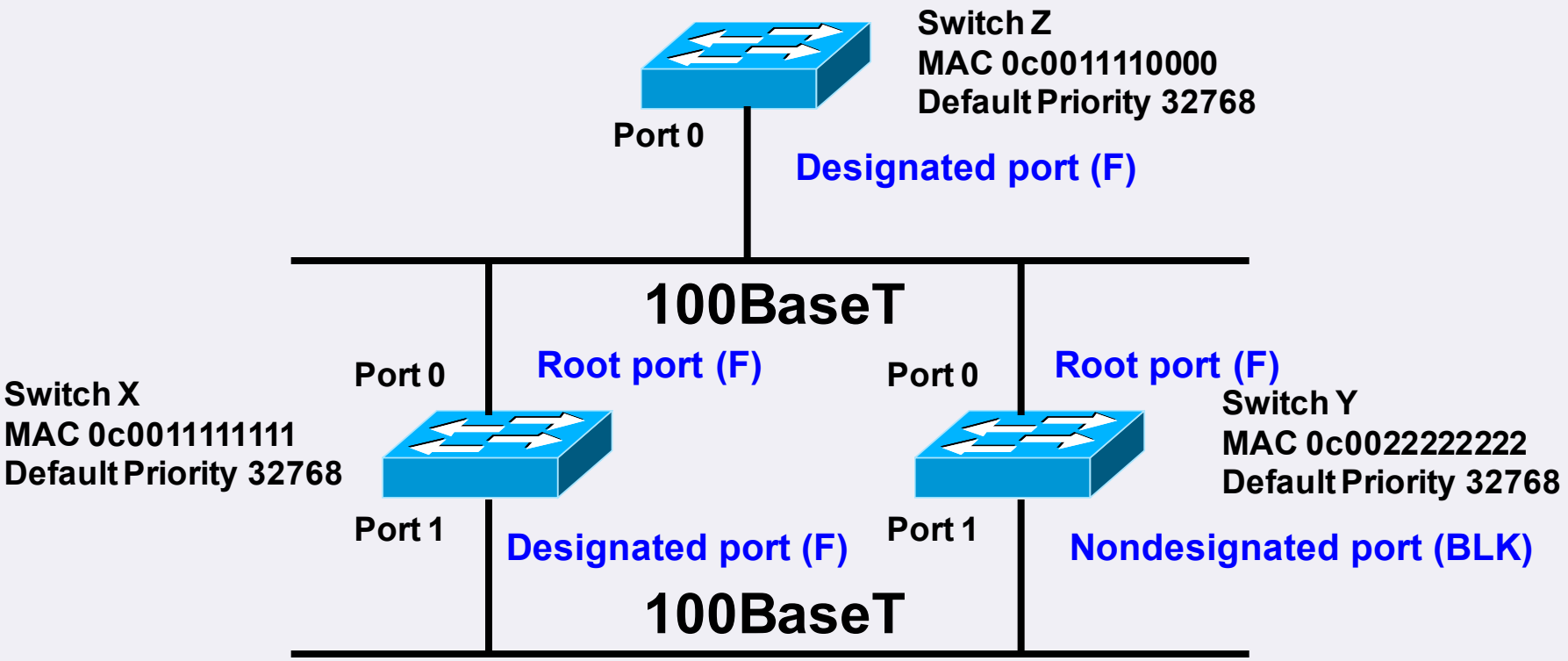


Spanning Tree Example



- Can you figure out:
  - What is the root bridge?
  - What are the designated, nondesignated, and root ports?
  - Which are the forwarding and blocking ports?

Spanning Tree Example (cont.)



- Can you figure out:
  - What is the root bridge?
  - What are the designated, nondesignated, and root ports?
  - Which are the forwarding and blocking ports?

STP Timers

Timer	Primary Purpose	Default
Hello Time	Time between sending of configuration BPDUs by Root Bridge	2 Secs
Forward Delay	Duration of listening and learning states	15 Secs
Max_Age	Time BPDU stored	20 Secs



The maximum waiting time for Hello BPDU before the switch decide the link is fail (~ 10 BPDU).

Spanning-Tree Port States

blocking → listening → learning → forwarding

Blocking - No frames forwarded, BPDUs heard

- All ports in a blocking state by default when switch starts.
- Used to prevent network loops.
- In this state it attempts to figure out which will port is to be the root port, which ports on the switch need to be the designated ports, and which ports will remain in a blocked state to break up any loops.
- Remains in this state for 20secs.

Listening - No frames forwarded, BPDUs heard but addresses not learnt

- Any designated port will now move to this state after 20secs, and any other ports remain in a blocked state.
- Listens to incoming Hello messages to ensure that there are no loops, but does not forward traffic or learn MAC addresses on the interface.
- Remains in this state for 15secs.

Learning - No frames forwarded, BPDUs heard and addresses learned

- Begins to process frames, populate its table, but does not forward the frames
- Remains in this state for 15secs.

Forwarding - Frames forwarded, BPDUs heard and addresses learned

- Sends and receives data frames on the bridge port.
- Forwarding ports have the lowest costs to the root bridge.
- A change in topology (a failed link, etc) causes the ports to enter listening and learning states.
- Typically switch ports are either in a **blocking or forwarding** state.

Disabled - No frames forwarded, no BPDUs heard

- special port state where the port is not participating in STP.

Verifying the STP Configuration & Operation

